

MASTER'S THESIS

Investor Embeddings from Institutional Holdings Spherical Cauchy Mixtures of Learned Portfolio Similarity, and Their Dynamics

submitted by

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Abstract

Gabaix et al. (2025) introduce asset embeddings: vector representations of assets and investors learned from institutional holdings by transformer encoders. Their investor-side model, Portfolio-Shares BERT (PS-BERT), is estimated and benchmarked, but the systematic classification of investors by investment strategy is left to future research, and its investor-side interpretation rests on nearest-neighbour lists narrated by a large language model. This thesis takes up that application: it replicates PS-BERT and extends it with model-based clustering on the sphere, using finite mixtures of spherical Cauchy distributions.

For the reference quarter (2025-Q4), an eight-component solution recovers economically identifiable populations: national and regional markets (China, continental Europe), a passive index product, distinct United States growth and income-and-real-assets groups, an Asian technology group, a cross-regional value group, and a diffuse residual population. Learned portfolio similarity thus tracks the investable universe—what an investor may hold—more strongly than investment orientation. Weighted pooling reorganises the partition substantially on identical inputs and ranks capitalisation above geography; reinstating the US-only restriction of Gabaix et al. (2025) post hoc reverses this, with clusters separating by capitalisation and value-growth-income orientation and the explanatory power of portfolio characteristics roughly tripling. Investment orientation is present in holdings-based embeddings but is a within-segment phenomenon, legible once the investable universe is held fixed, which reconciles these results with the strategy-like groupings reported in the source paper.

A further extension asks how investors move over time. Annual cross-sections are aligned by orthogonal Procrustes, anchored on the investors whose disclosed holdings changed least; this places unchanged portfolios more accurately, out of sample, than aligning on the whole population. In the aligned space, displacement rises with portfolio turnover in every one of twenty annual steps and under every anchor rule, so what is measured is movement of the portfolio rather than of the coordinate system. Applying the reference-quarter mixture to every aligned year shows the clusters to be durable populations: 64 % of investors keep their component from one year to the next and 46 % still share it a decade later, against 17 % under chance. Re-estimating the mixture year by year, however, recovers the reference partition only moderately, so the clusters are durable as populations of investors yet fragile as objects to re-estimate. Investors reallocate at identifiable moments—notably 2008, 2015–2017 and 2020–2021.

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1 Introduction

1.1 Motivation

Every quarter, thousands of institutional investors disclose what they own. A portfolio is not a number; it is a long, sparse, changing list. The conventional response has been compression through predefined categories: an investor is “large-cap growth”, holds “25 % technology”, tracks “the S&P 500”. Such labels are convenient, but they are exogenous. They describe portfolios in a vocabulary chosen before looking at the data, and they presuppose that the dimensions along which investors genuinely differ are the dimensions someone thought to name.

A recent line of work in financial economics proposes to replace hand-picked vocabularies with *learned representations*. Gabaix et al. (2025) show that the toolkit of natural language processing—word embeddings, transformers—transfers naturally to holdings data once one notices a structural analogy: as documents are sequences of words, portfolios are collections of assets, and the information about an asset is carried by *who* holds it, just as the meaning of a word is carried by the company it keeps. Embedding methods compress the joint holding structure into vectors in which geometric proximity encodes similarity.

1.2 The starting point: an unexploited investor side

Gabaix et al. (2025) estimate two models. Ownership-Shares BERT (OS-BERT) reads each asset as a sequence of its owners and yields *asset* embeddings; Portfolio-Shares BERT (PS-BERT) reads each portfolio as a sequence of its holdings and yields *investor* embeddings. The paper’s empirical weight falls almost entirely on the asset side, where the embeddings are validated against firm characteristics, return comovement and relative valuations. The investor side is estimated, shown to perform well on a masked-position benchmark, and then set aside: the paper states explicitly that investor embeddings are not fully explored, and lists three applications left for future research—systematic classification of investors by investment strategy, performance evaluation, and the detection of crowded trades.

Where investor embeddings are interpreted, the method is illustrative rather than systematic. For a focal fund, the nine or ten closest investors by cosine similarity are listed, their prospectus strategy descriptions are fed to a large language model, and the model is asked to name what they have in common. The demonstration is convincing—eight of the nine investors closest to a small-cap value ETF have “small cap value” in their name—but it is a just demonstration. It examines a handful of neighbourhoods rather than the whole space.

A second opening appears in the source paper’s own list of possible refinements. PS-BERT uses *ranked* positions: holdings are ordered by portfolio share, and the share itself is then discarded. The paper notes that dollar magnitudes could instead enter the objective function directly, and leaves the question open. The aggregation step—how a set of token vectors becomes one investor vector—inherits the same property: the source specification averages

the contextual embeddings of an investor’s largest 62 positions without weights, so a thirty-percent position and a thirty-basis-point position count equally.

This thesis takes up the first application on the source paper’s list, using the second opening as one of its instruments, and adds a question the source framework does not pose. Because a model is estimated separately for every quarter, nothing in that framework asks whether the investor structure it recovers persists—whether the groupings are durable populations or an artefact of the cross-section in which they were estimated.

1.3 Research question

The question has two halves:

Can investor embeddings learned from portfolio holdings alone recover clusters of investors that are economically distinct—and are those clusters stable objects over time, or do investors reallocate between them at identifiable moments?

The first half is deliberately open about what “distinct” will turn out to mean. It does not presuppose that the answer is investment style, or geography, or size; it asks what the data volunteer when nothing is imposed, and then requires the answer to be demonstrated. Answering it has three parts. The geometry must contain structure at all. That structure must be partitioned into populations. And those populations—which the geometry leaves anonymous, being no more than directions in a 64-dimensional space—must be given economic content by confronting them with attributes measured independently of the embedding, and those attributes ranked against one another.

The second half asks what the first cannot. A partition of one quarter is compatible with clusters that are durable features of the market and with components that are reassembled differently every time the model is estimated, and the two have very different implications for what the classification is worth. Because the source model is trained separately in each quarter, its embedding spaces are not comparable across time, so components must be tracked by using orthogonal Procrustes.

1.4 What is replicated and what is extended

Because the argument of this thesis depends on which parts of the pipeline are inherited and which are new, the two are separated explicitly at the outset. Table 1 is the map; it is referred to throughout.

The extensions are not independent contributions bolted together; they stand in a definite relation to one another. The mixture model and the characterisation procedure are what make the source paper’s postponed application—systematic investor classification—executable at all: without a partition there is nothing to characterise, and without measurement any characterisation is merely asserted. The global universe and the weighted aggregation rule

are then the two levers that the resulting machinery can be used to pull, and each turns out to change the answer: weighting changes which investors are judged similar, and the breadth of the universe, tested by reinstating the source restriction in Section 5.7, changes which dimension leads. The cross-quarter alignment of Chapter 6 then applies the finished machinery to the second half of the research question, and inherits from the others the constraints that make it tractable: a fixed K , a partition rather than a neighbourhood, and attributes with which persistence can be interpreted.

1.5 Contributions

The thesis makes five contributions.

First, it provides an end-to-end, reproducible pipeline—from raw WRDS extracts to characterised components—for representing institutional investors as spherical embeddings, implemented in R and Python and designed to run per quarter over an 85-quarter global panel.

Second, it isolates the aggregation rule as an object of study, showing that replacing unweighted with weighted pooling reorganises the partition substantially while leaving a stable core intact—a result about how portfolio aggregation shapes learned investor representations that the literature has not previously examined, and a first step toward the magnitude-aware objective Gabaix et al. (2025) leave open.

Third, it transfers the spherical Cauchy mixture methodology of Sablica et al. (2025) to a financial dataset and documents how likelihood-based model selection behaves on such data.

Fourth, it develops a procedure for making the economic content of an embedding-based clustering measurable instead of merely assertable: a fixed set of descriptive dimensions applied uniformly to every component, and an association ranking that scores each attribute against component membership. This is the direct replacement for the narrative interpretation of the source paper, and it is what allows the central finding—that the components are organised principally by the investable universe—to be stated as a ranking rather than an impression.

Fifth, it extends the analysis from one cross-section to the panel, aligning per-quarter partitions through orthogonal procrustes and measuring both component persistence and the timing of investor reallocation. The source framework, estimating one model per quarter, provides no way to compare embedding spaces across time; the alignment developed here supplies one.

1.6 Outline

Chapter 2 reviews the three literatures this work sits between, beginning with a detailed account of the framework being extended. Chapter 3 documents the data and its trans-

formation into model inputs, marking at each step whether the choice is inherited or new, and introduces the attribute tables the characterisation requires. Chapter 4 presents the replicated embedding model first and then each extension in turn. Chapter 5 contains the empirical analysis for the reference quarter: the replicated arm, then the weighted arm, then their comparison, then the reinstatement of the source paper's universe. Chapter 6 extends the analysis across the panel, aligning per-quarter partitions and asking whether the components persist and when investors move between them. Chapter 7 discusses and concludes, separating what replicates from what the extensions add.

Table 1: Replication and extension map. The baseline arm reproduces the investor-side pipeline of Gabaix et al. (2025); the remaining rows mark where this thesis departs, each departure isolated so that its effect on the results can be measured separately.

Stage	Gabaix et al. (2025)	This thesis	Status
Holdings source	FactSet Ownership via WRDS; fund-level data for registered vehicles, 13F filings restricted to hedge funds	Identical	replicated
Sample filters	Top position $\leq 75\%$; iterate ≥ 20 stocks per investor and ≥ 20 investors per stock to convergence	Identical	replicated
Security universe	US stocks only, via a merge to CRSP / Compustat; micro- and nano-caps removed	No merge; the global universe as reported	extended
Sample period	2005-Q1 to 2022-Q4	2005-Q1 to 2026-Q1 (85 quarters)	extended
Sequence construction	Holdings sorted by portfolio share; issuer tokens; 62-token context window	Identical, plus position weights and an explicit chunk index retained alongside each token	replicated
Encoder and training	Per-quarter BERT; masked-token pretraining; contrastive fine-tuning on odd/even halves of the same portfolio	Identical	replicated
Investor embedding	Unweighted mean of contextual token vectors over the largest 62 positions	Baseline arm identical; second arm weights each position by its portfolio share	extended
Grouping of investors	Nearest neighbours of a focal investor by cosine similarity; no partition	Finite mixture of spherical Cauchy distributions on $\mathbb{S}^{63}$; soft assignment; likelihood-based model selection	extended
Interpretation	Prospectus strategy text for ten neighbours, summarised by an LLM	Structured FactSet attributes on four dimensions, plus an association ranking scoring every attribute against component membership; cluster labels condense these tabulated attributes	extended
Treatment of time	One model per cross-section; embeddings interpreted within a quarter, with no cross-quarter identification	Per-quarter partitions aligned, giving component persistence and investor reallocation over time	extended

2 Related Work

2.1 The framework being extended: asset embeddings

2.1.1 The analogy and the two models

Gabaix et al. (2025) coin the term *asset embeddings*: vector representations of assets and, dually, investors, learned from institutional holdings in analogy to word embeddings learned from text corpora. Their central observation is that holdings data possess the same bipartite “who-appears-with-whom” structure that makes distributional semantics work: an asset is characterised by the portfolios containing it, an investor by the assets held.

The analogy is implemented twice, in mirror image. Ownership-Shares BERT (OS-BERT) treats an asset as a sequence of its owners, ordered by ownership share, and produces asset embeddings. Portfolio-Shares BERT (PS-BERT) treats a portfolio as a sequence of its holdings, ordered by portfolio share, and produces investor embeddings. Both are trained in two stages: masked-token pretraining in the manner of Devlin et al. (2019), followed by contrastive fine-tuning in the manner of Reimers and Gurevych (2019), in which the even- and odd-ranked halves of the same sequence form a positive pair, so that cosine similarity becomes a valid distance measure on the resulting vectors. An embedding is then the average of the contextual token vectors over the largest 62 elements, the length of the context window.

2.1.2 What the source paper establishes, and what it leaves open

The asset side carries the paper. Asset embeddings are shown to explain relative valuations, return comovement and observed firm characteristics, and to outperform recommender-system and Word2Vec baselines on a managed-portfolio benchmark. The investor side is estimated and validated on a masked-position benchmark, but is otherwise held in reserve: the paper states that investor embeddings are not fully explored and identifies three applications for future research, of which the first is the systematic classification of investors by investment strategy—the observation being that standard investor characteristics such as institutional type, size and activeness explain little of the variation in strategies.

Where investor embeddings are interpreted, the vehicle is text. For a focal fund, the closest investors by cosine similarity are listed and their prospectus strategy descriptions are passed to a large language model with a prompt asking for the commonalities. The results are persuasive as illustrations. They are, however, neighbourhood-local, unranked against alternative explanations, and dependent on a text corpus—US summary prospectuses, FactSet strategy paragraphs, fund websites—whose coverage outside the United States is uneven.

Finally, the paper is explicit that PS-BERT uses ranked rather than magnitude-weighted positions, and lists the incorporation of log dollar holdings into the objective as a possible refinement, alongside an integrated model of both sides.

2.1.3 Where this thesis departs

This thesis adopts the source paper’s sequence construction on the investor side—the ordering of holdings by weight, the 62-token context window, the two-stage training procedure—and its sample-construction filters verbatim. It departs in the respects tabulated in Table 1, and it is worth stating why each departure is substantive rather than cosmetic.

Removing the CRSP merge changes the object being learned. In a US-only universe, two funds that hold no security in common are unusual; in a global universe they are the norm, because eligibility rather than preference separates them. Section 5.7 shows that this single difference accounts for the divergence between the investor groupings reported here and those reported in the source paper.

Weighting the pooling operator addresses at the aggregation stage the magnitude question the source paper raises at the objective stage. It is a weaker intervention than retraining against dollar holdings, but it has the compensating virtue of being estimable from identical inputs, so the comparison isolates aggregation exactly.

Replacing neighbourhood inspection with a mixture model turns qualitative similarity into a partition of the entire population, and adds two quantities the source approach does not produce: a per-component concentration parameter and posterior membership probabilities.

Replacing LLM narration with structured attributes and an association ranking makes the interpretation falsifiable. A narrative account of why a group of investors resembles one another cannot be compared against the account it displaced; a Cramér’s V can.

2.2 Institutional holdings in empirical finance

Quarterly disclosure of institutional portfolios—13F filings in the United States and their fund-level counterparts elsewhere—has supported two broad research programmes. One treats holdings as a record of *behaviour*, asking what institutions do and whether they do it together. The other treats holdings as a lens on *prices*, asking what the ownership structure of a security implies about how it will be traded and valued. This thesis sits squarely in the first. What both have in common, and what motivates the representation learned here, is that they require a notion of when two investors are alike—and have generally had to impose one in advance.

Correlated trading. The herding literature asks whether institutions buy and sell the same securities at the same time. Lakonishok et al. (1992) introduced the measure that organised the field, comparing the fraction of institutions trading a stock on the same side against what independent trading would produce, and found the effect small in a sample of pension funds. Wermers (1999) found it stronger among mutual funds, particularly in small stocks and in growth-oriented funds, and documented that herded-on stocks subsequently adjust in the direction of the herd. Sias (2004) reframed the question dynamically, showing that

institutional demand is strongly autocorrelated across quarters and that institutions follow one another rather than merely responding to common signals. The measures share a structural feature: they are computed per security, on trades, and within a group of investors defined beforehand—all institutions, or all funds of a declared type. Whether the group is the right one is not something the measure can answer.

Ownership structure and prices. A second strand asks what a security's ownership implies for its price dynamics. Coval and Stafford (2007) showed that funds facing large outflows liquidate existing positions rather than rebalance, and that securities widely held by such funds experience price pressure and subsequent reversal. Greenwood and Thesmar (2011) formalised the exposure this creates: their fragility measure is a quadratic form in ownership weights, in which the covariance of investors' liquidity shocks replaces the independence a concentration index would assume, so that a security is fragile when it is held by few investors, or by investors whose flows move together. The construction is the closest structural analogue in the literature to a similarity-based reading of ownership, and it is instructive that it too requires an exogenous notion of when two investors move together—there, the covariance of their flows. Antón and Polk (2014) apply the idea across securities rather than within one, using shared mutual-fund ownership to link pairs of stocks and showing that the links predict return comovement and cross-stock return predictability.

Crowded trades. The term itself belongs to a smaller and less settled literature, in part because the object is hard to define. Khandani and Lo (2011) reconstruct the August 2007 quant episode as the unwinding of positions held simultaneously by many quantitative market-neutral funds, an event visible in factor returns without being visible in any single fund's disclosure. Stein (2009) gives the conceptual account: when arbitrageurs cannot observe how much capital is already deployed against an anomaly, crowding and the disorderly unwinds it produces are equilibrium outcomes rather than mistakes. Pojarliev and Levich (2011) propose a detection method for currency managers based on common factor exposures. Across these, the recurring difficulty is the same one the herding literature faces: identifying crowding requires knowing which investors count as similar, and the answer is usually proxied by a declared style, a factor loading, or a self-reported strategy.

Style classification and its limits. That proxy has been questioned for as long as it has been used. Sharpe (1992) inferred style from return exposures rather than declared mandates precisely because the two need not agree; Brown and Goetzmann (1997) classified funds directly from their returns and found groupings that cut across the categories the industry assigns, with the data-derived classes explaining performance dispersion better than the declared ones. Chan et al. (2002) document substantial drift between a fund's stated style and the characteristics of what it actually holds, and Cremers and Petajisto (2009) construct a holdings-based measure of activeness that separates funds a categorical scheme treats as identical.

Demand systems. A more recent literature models institutional holdings structurally. Kojen and Yogo (2019) treat each investor as a demand curve over securities, with demand a function of a small set of observed characteristics plus a latent residual, and aggregate these curves to obtain equilibrium prices; Kojen et al. (2024) use the framework to attribute variation in valuations to particular investor groups. The approach is parametric by construction: what an investor is, is a vector of coefficients on pre-selected characteristics, and everything unexplained falls into latent demand. Embeddings offer the complementary move, learning the description rather than specifying it, at the cost of a representation whose axes have no economic labels—which is the problem Chapter 5 is built to address.

Learned similarity as a substitute for fixed categories. The strategy of replacing an imposed classification with one estimated from data has a direct precedent outside the investor setting. Hoberg and Phillips (2016) replace fixed industry codes with a similarity network built from the text of firms’ product descriptions, and show that the resulting industries are time-varying, firm-specific and better at explaining competitive outcomes than the codes they displace. The asset-embeddings framework of Section 2.1 is the analogous move for portfolios, with co-holding playing the role of shared vocabulary.

Where this thesis sits. Two gaps in the literature above define the space this thesis occupies. The first is the one Gabaix et al. (2025) name and leave open: the systematic classification of investors by what they hold. Every herding, crowding and style measure surveyed here requires a prior judgement about which investors are alike, and takes it from an institutional type, a declared style, or a factor exposure—all exogenous to the holdings they are used to interpret. Chapters 4 and 5 construct that judgement from the holdings themselves and then ask what it turns out to encode.

The second gap is temporal. The literature on institutional behaviour is overwhelmingly about *change*—herding is a statement about correlated trades, fire sales about forced liquidation, style drift about mandates that move—yet the embeddings literature it would draw on is resolutely cross-sectional, estimating a representation for one quarter and interpreting it there. Chapter 6 closes that gap for the investor side, asking whether the populations recovered in one quarter are present in the next and when their membership turns over.

2.3 Text embeddings and transformer encoders

The modelling machinery originates in natural language processing. BERT (Devlin et al., 2019) established masked-token pretraining of bidirectional transformer encoders as a general-purpose representation learner. Sentence-BERT (Reimers and Gurevych, 2019) showed that useful *sequence-level* representations require an explicit pooling choice, with mean pooling over token embeddings a robust default—a precedent followed both by Gabaix et al. (2025) and in the baseline specification here, since the alternative CLS-token summary only acquires meaning under training objectives (such as next-sentence

prediction) that neither pipeline uses. SimCSE (Gao et al., 2021) demonstrated that a contrastive fine-tuning stage with an InfoNCE objective substantially improves the geometry of sentence embeddings; the fine-tuning stage used here is its portfolio analogue, with two disjoint halves of the same portfolio playing the role of two views of the same sentence.

2.4 Clustering on the sphere

This is the literature the source paper does not draw on, and from which the mixture model of this thesis is taken. Once representations are L^2 -normalised, their natural home is the unit sphere $\mathbb{S}^{d-1}$, and Euclidean clustering methods are geometrically misspecified. The spherical k -means algorithm of Hornik et al. (2012) replaces squared Euclidean distance with cosine dissimilarity. Its model-based generalisation replaces prototypes with component distributions on the sphere: mixtures of von Mises–Fisher distributions (Banerjee et al., 2005; Hornik and Grün, 2014) are the classical choice, while Sablica et al. (2025) introduce mixtures of Poisson kernel-based and spherical Cauchy distributions (Golzy and Markatou, 2020; Kato and McCullagh, 2020), implemented in the **circlus** package as drivers for the EM framework of **flexmix** (Leisch, 2004; Dempster et al., 1977). The spherical Cauchy family has two properties valuable at the present scale: heavier tails than the von Mises–Fisher, granting robustness to the outlying portfolios that institutional data inevitably contain, and a density requiring no special functions, so that estimation reduces to matrix arithmetic. This thesis is, to the author’s knowledge, among the first applications of the **circlus** methodology outside its introducing paper, and the first to apply it to holdings-based embeddings.

3 Data

The sample construction of this section follows Appendix D of Gabaix et al. (2025) step by step, with two deliberate departures: the security universe is not restricted to the United States, and the token sequences retain position weights and a chunk index so that the weighted specification is estimable from identical inputs. Each section below states which case applies.

3.1 Source and coverage

Holdings are drawn from the FactSet Ownership database, accessed through Wharton Research Data Services (WRDS). Two source tables are combined, exactly as in the source paper. Hedge fund holdings come from the 13F filings, restricted to filers whose entity sub-type is HF, within a ± 7 day window of the quarter end. Holdings of registered funds come from the fund-level table, joined to the fund entity table for the fund type and restricted to open-end funds, exchange-traded funds, closed-end funds and variable annuity portfolios, within a ± 5 day window; where a fund reports more than once in the window, the most recent report is kept.

Using fund-level rather than filer-level data for registered vehicles follows Gabaix et al. (2025)—who report that the more granular data produce better embeddings—and matters for what a “portfolio” means: the unit of observation is an individual fund with its own mandate, not the consolidated book of an asset manager. It also explains why several components identified in Section 5.5 are extremely concentrated geographically—a single-country equity fund is a normal object in this population, whereas a consolidated manager would not be.

The restriction on the 13F side deserves emphasis because it defines the population under study. Of the institutions that file 13F reports—banks, insurers, pension funds, endowments and hedge funds—only hedge funds are retained. The universe analysed here is therefore “registered funds plus hedge funds”. This too is inherited.

The departure. The source pipeline merges the holdings to the CRSP–Compustat link file by date and historical CUSIP, keeps US firms with non-missing characteristics, and drops micro- and nano-caps. That merge is omitted here. Securities are identified at the FactSet issuer-entity level and no geographic or exchange restriction is imposed, so the universe is whatever institutional investors report holding worldwide. The consequence is not incidental: it admits precisely the variation—across national markets and mandates—that Chapter 5 finds to dominate the geometry, and it is the reason Section 5.7 can reinstate the source restriction as a controlled comparison rather than having to argue about it.

The panel spans 85 quarters, 2005-Q1 through 2026-Q1, against the source paper’s 2005-Q1 to 2022-Q4. The number of investors surviving the filters grows from 9,142 in the earliest

quarter to 32,639 at its peak, with a median of 19,232, and displays a pronounced seasonal pattern: averaged over the panel, second- and fourth-quarter counts (21,663 and 22,194) exceed first- and third-quarter counts (16,811 and 17,098) by 29.3%. The regularity is a feature of the disclosure calendar since there are investors that report only semi-annually. As the descriptive statistics below set out, it is innocuous for a single-quarter analysis but must be handled explicitly in the panel analysis of Chapter 6, where it determines which quarters may be compared at all. The reference quarter used throughout Chapter 5 is 2025-Q4.

3.2 Filtering

The raw ownership tables mix reporting conventions, contain stale and duplicate filings, and include portfolios too concentrated or too thinly held to carry co-holding information. Two filters are applied independently to each quarter. Both are taken unchanged from Gabaix et al. (2025), including the numerical thresholds, so that any difference in results cannot be attributed to sample cleaning.

Concentration cap. Investor-quarters whose largest single position exceeds 75 % of portfolio value are dropped. Such portfolios are typically holding vehicles or single-stake entities; their embedding would be determined almost entirely by one token.

Bipartite pruning. The holdings data form a bipartite graph between investors and issuers, and both sides must be dense enough for the masked-token objective to have signal. Investor-quarters with fewer than 20 positions are removed, as are issuer-quarters held by fewer than 20 investors. Because each removal can push the other side below its threshold, the two filters are applied repeatedly until a fixed point is reached.

One consequence of the wider universe shows up here. The source paper removes micro- and nano-caps explicitly, using the size categories of the firm-characteristics dataset it merges to. With no merge available, no such filter is applied, and none is needed: micro- and nano-cap issuers are held by too few investors to survive the 20-holder rule, so the bipartite pruning does the same work endogenously.

3.3 From portfolios to token sequences

For each investor-quarter, holdings are sorted by market value in descending order and mapped to a sequence of issuer-identifier tokens. Following Gabaix et al. (2025), the sequence is the portfolio analogue of a sentence. Because the encoder’s context window is 62 tokens, portfolios longer than 62 positions are split into equal size consecutive chunks; all chunks are used for pretraining, so the masked-token objective visits the tail of large books, while the investor embedding is computed from the largest 62 positions, as in the source specification.

The one addition. Each token is stored together with the share of portfolio value it represents. This is a departure from the source pipeline, where magnitudes are used to order the sequence and then discarded, and it is what makes the weighted specification of Section 4.2 estimable from exactly the same inputs as the baseline. Chunks additionally carry an explicit index, so that a position's rank within the whole portfolio is recoverable rather than inferred from file order—which matters because the weights are normalised across the full portfolio, so a mis-ordered chunk would assign the largest weights to tail positions.

3.4 Attributes for characterisation

The token sequences are deliberately impoverished: an investor is a list of identifiers, an issuer is an identifier. Nothing in them says who the investor is, where the issuer is domiciled, or what either does. Clustering such sequences produces groups that can be described geometrically but not economically.

This is the point at which Gabaix et al. (2025) turn to text: prospectus strategy descriptions, fed to a language model. This thesis substitutes structured data. Two reference tables are assembled from the FactSet entity and ownership tables, one per side of the bipartite graph, and reused by every subsequent analysis. The substitution buys three things: coverage that does not depend on the availability of English-language prospectuses, attributes that are the same for every component and therefore comparable across them, and quantities that can be scored.

The **investor table** carries, for each investor: name, domicile and entity type from the FactSet entity master; fund type, family, declared investment objective (specialisation, asset type, region and country of mandate) and a block of portfolio characteristics—beta, price-earnings and price-book ratios, dividend yield, sales growth, price momentum and relative strength—from the fund tables; declared style, turnover classification and assets under management; the identity of the managing institution and the ultimate parent entity; and panel statistics computed locally, such as the quarters in which the investor appears and its median portfolio size.

The **security table** carries, for each issuer: name, domicile, sector and industry classification; the primary listing's exchange, issue type and capitalisation group; market capitalisation; cross-identifiers; and the number of investors holding it.

In the reference quarter, 99 % of investors match the investor table and 100 % of held issuers match the security table, so the characterisations that follow describe substantially the whole sample.

One caveat is carried forward. The style, turnover and characteristic fields are *current* snapshots rather than histories: they describe the investor as classified at extraction time. For the single-quarter analysis of Chapter 5, the reference quarter is a close approximation. For the panel analysis of Chapter 6 it would not be, since a fund that changed mandate would

Table 2: Panel summary, 2005-Q1 to 2026-Q1 (85 quarters). Portfolio size is the number of positions held by an investor in a quarter.

Quarters	85
Investors per quarter: minimum	9,142
median	19,232
maximum	32,639
Issuers per quarter (median)	11,362
Portfolio size (median across the panel)	53
Share of investors above the 62-token window	42.6 %

carry its final label backwards through the sample.

3.5 Descriptive statistics

Table 2 summarises the panel. Two features bear on the modelling choices that follow.

First, portfolio sizes are strongly right-skewed. In the reference quarter the median investor holds 60 positions and the quartiles are 38 and 124, but the mean is 173.9 and the largest book holds 9,195 positions. The context window of 62 tokens—inherited from the source specification—therefore sits close to the median rather than in the tail: 47.8 % of investors in the reference quarter, and 42.6 % across the panel, hold more positions than the encoder can see at once.

Second, the investor count is seasonal rather than monotone, as noted in Section 3.1. Any analysis comparing adjacent quarters must account for this, since roughly a quarter of the sample enters and leaves on a six-month cycle for reporting reasons unconnected to investor behaviour. For this reason the analysis Chapter 6 is done using only the last quarter of each year.

Figure 1 shows both series over the panel and makes the two features visible at once.

The growth is substantial: annual mean investor counts roughly double between 2005 (11,548) and 2016 (21,656), pause through 2018, and resume, reaching 27,810 in 2025. Distinct issuers held rise from 6,488 to 16,792 over the same period. Exchange-traded funds grow from an annual mean of 166 investor-quarters in 2005 to 6,144 in 2025, so a growing share of the sample consists of vehicles whose books are determined by an index rather than by a manager, which is worth keeping in mind when reading the passive components of Section 5.5.

The sawtooth is the more consequential feature for what follows, and the right-hand panel isolates it. Averaged over the panel, second- and fourth-quarter investor counts exceed first- and third-quarter counts by 29.3 %, and the alternation is exact: every second and fourth quarter of the panel is a local maximum and every first and third a local minimum, without exception.

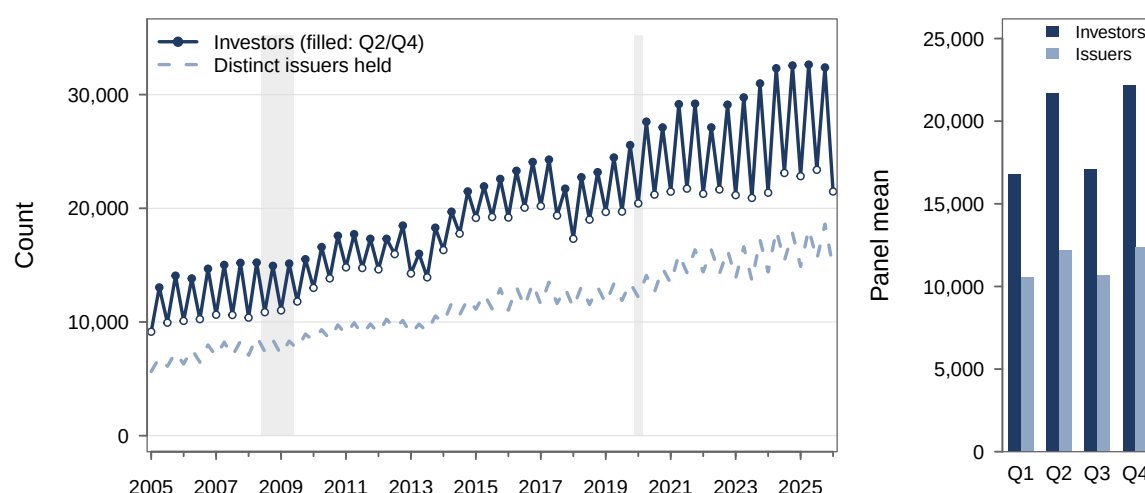


Figure 1: Panel coverage, 2005-Q1 to 2026-Q1. *Left*: the number of investors (solid) and of distinct issuers held (dashed) surviving the filters of Section 3.2 in each quarter. Markers on the investor series are filled for second and fourth quarters and hollow for first and third. Shaded bands mark 2008-Q3–2009-Q2 and 2020-Q1. *Right*: the panel mean of each series by calendar quarter, investors in dark and issuers in light.

The explanation is the disclosure calendar. Holdings for registered vehicles are recorded when a fund publishes them, and a large part of the non-US fund universe discloses its portfolio only twice a year, on a financial year end that is December for the large majority—so such a fund appears in the June and December windows and is simply absent from the March and September ones. Nothing is carried forward or interpolated, so each quarter counts the investors who filed. United States registered funds report at least quarterly, as do the hedge funds retained here, and the panel is therefore a mixture of two reporting regimes.

Two features of the data support that reading. The gap is close to constant in proportional terms in every year of the sample, even though the annual mean investor count more than doubles between 2005 and 2025, which is difficult to reconcile with any account based on fund launches or on broadening vendor coverage. And the issuer series shows the same cycle at roughly half the amplitude (15.5 %): the funds that enter in June and December hold, for the most part, securities that were already held by somebody in March and September, so the influx enlarges the investor side of the bipartite graph without much enlarging the issuer side. Part of even that residual amplitude is induced, since a larger investor side pushes more issuers over the 20-holder threshold of Section 3.2.

4 Methodology

This section is ordered by Table 1. Section 4.1 documents the replicated PS-BERT pipeline, including the points at which the source paper leaves a hyperparameter unspecified and a choice had to be made here. Sections 4.2 to 4.5 present the extensions in turn: weighted pooling, the spherical mixture, and the characterisation and association procedure.

4.1 The replicated baseline: PS-BERT

4.1.1 Architecture and training

The encoder is a deliberately small BERT-style transformer: 4 layers, 2 attention heads, hidden dimension 64, and a context window of 62 holding tokens plus special tokens. The vocabulary consists of the issuer identifiers appearing in the quarter. The small footprint reflects both the modest vocabulary relative to natural language and the need to train one model per quarter across an 85-quarter panel on a single GPU—the per-cross-section design being itself inherited, since Gabaix et al. (2025) train one model per quarter in their baseline specification and report only a limited gain from pretraining on longer histories.

Training follows the masked-language-model recipe of Devlin et al. (2019): 15 % of tokens in each portfolio sequence are masked, of which 80 % are replaced by the mask token, 10 % by a random issuer and 10 % left unchanged, and the model is trained to recover them from the surrounding holdings. The task forces the encoder to internalise which issuers plausibly appear together in a book.

A second stage shapes the investor-level geometry. Each portfolio is split into two disjoint halves (even- and odd-ranked positions), the halves are encoded separately, and an InfoNCE objective with in-batch negatives pulls the two views of the same investor together while pushing apart views of different investors. The implied invariance is economically sensible: two random halves of one book should identify the same investor. It is also exactly the fine-tuning construction of the source paper, whose purpose there is likewise to make cosine similarity a valid distance measure on the resulting vectors—a precondition for everything the mixture model does with them.

4.1.2 Replication fidelity

A replication is only as informative as its account of what could not be replicated exactly. Table 3 separates the settings the source paper specifies from those it does not.

The logit scale deserves particular note, since it is the one unspecified value with a direct effect on the geometry that is then clustered. At $s = 20$ the contrastive objective sharpens the angular separation between investors considerably; a smaller value would flatten it. The choice follows NLP precedent rather than anything in the source paper, and any statement in this thesis about the scale of cosine similarities should be read with that in mind.

Table 3: Replication fidelity. “Specified” settings are stated in Gabaix et al. (2025); “unspecified” ones were chosen here and are implementation choices, not inherited ones.

Setting	In the source	Value used here
Sequence ordering	specified	portfolio share, descending
Context window	specified	62 holding tokens
Masking rate and 80/10/10 split	specified	15 %, 80/10/10
Positive pairs for fine-tuning	specified	even- vs. odd-ranked halves
Similarity measure	specified	cosine
Pooling operator	specified	unweighted mean over the largest 62 positions
Training frequency	specified	one model per cross section
Embedding dimension	varied (4–128)	64
Layers, attention heads	unspecified	4 layers, 2 heads
Contrastive logit scale	unspecified	$s = 20$, i.e. temperature $\tau = 0.05$, the SimCSE default (Gao et al., 2021)
Epochs, batch size, schedule	unspecified	10 MLM epochs, 3 fine-tuning epochs, batch 64, AdamW with cosine schedule and 10 % warmup
Train/validation split	unspecified	90/10 by investor, seed 42

4.1.3 Training diagnostics

Table 4 reports the training diagnostics for the five quarters estimated under both specifications. Three points follow.

First, pretraining works but does not come close to solving the masked task, which is the expected and desirable outcome. The loss falls from around 8.9 to around 6.05 against a uniform-guess reference of 9.6 to 9.9, so the model recovers a substantial part of the available signal while remaining far from reconstruction—unsurprising when each mask has some fifteen thousand candidate issuers. Validation loss sits marginally *below* training loss in every quarter, within 0.06, which is the usual consequence of comparing a dropout-active training average against a dropout-free evaluation pass; in no quarter does the gap open in the direction that would indicate overfitting.

Second, the masked-token losses of the two specifications are not merely close but identical, to every digit recorded, in all ten pretraining epochs of all eighty-five quarters. This is an internal check: the masked-token objective operates on tokens and never invokes the pooling operator, so with a common seed the two specifications execute the same computation at this stage. The two arms therefore enter fine-tuning from the same encoder, and any subsequent difference between them is attributable to aggregation alone, as designed.

Third, the contrastive losses differ systematically. At the first epoch the weighted specification is behind in every quarter (around 0.47 against 0.40 in Table 4); by the final epoch it has overtaken the baseline (around 0.15 against 0.16). The ordering is stable well beyond these five quarters: across all eighty-five quarters from 2005-Q1 to 2026-Q1, the weighted specification ends below the baseline in eighty-three, with a mean gap of -0.013 and a standard

Table 4: Training diagnostics. The random baseline is $\log(V)$, the cross-entropy of a uniform guess over a vocabulary of V issuers, and is the reference against which the fitted masked-token loss should be read. Fine-tuning losses are not strictly comparable across specifications, because the pooling operator enters the objective itself.

Spec.	Quarter	Vocab.	Masked-token loss				Contrastive	
			random	first	last	val.	first	last
Base	2025-Q1	15,022	9.62	8.85	6.07	6.05	0.425	0.179
Base	2025-Q2	18,442	9.82	8.95	6.02	5.96	0.328	0.124
Base	2025-Q3	15,706	9.66	8.87	6.10	6.04	0.408	0.176
Base	2025-Q4	18,894	9.85	8.98	6.04	6.01	0.334	0.125
Base	2026-Q1	14,993	9.62	8.86	6.05	6.00	0.401	0.178
Wtd.	2025-Q1	15,022	9.62	8.85	6.07	6.05	0.496	0.160
Wtd.	2025-Q2	18,442	9.82	8.95	6.02	5.96	0.376	0.117
Wtd.	2025-Q3	15,706	9.66	8.87	6.10	6.04	0.481	0.161
Wtd.	2025-Q4	18,894	9.85	8.98	6.04	6.01	0.381	0.114
Wtd.	2026-Q1	14,993	9.62	8.86	6.05	6.00	0.473	0.162

deviation of 0.008.

4.1.4 Pooling and normalisation

An investor’s embedding is the mean of its token embeddings, L^2 -normalised to unit length, following the Sentence-BERT precedent (Reimers and Gurevych, 2019) as the source paper does. Writing $h_{i1}, \dots, h_{iL_i}$ for the contextual token vectors of investor i ’s sequence,

$$x_i^{\text{base}} = \frac{\bar{h}_i}{\|\bar{h}_i\|}, \quad \bar{h}_i = \frac{1}{L_i} \sum_{\ell=1}^{L_i} h_{i\ell}, \quad L_i = \min(n_i, 62), \quad (1)$$

where n_i is the number of positions in the portfolio. The output per quarter is a matrix $X \in \mathbb{R}^{n \times 64}$ with unit-norm rows: n points on $\mathbb{S}^{63}$, where the cosine of the angle between two rows measures portfolio similarity.

Equation (1) embodies a substantive choice that is easy to overlook, and it is the choice the weighted specification interrogates. The average is *unweighted*: a position representing thirty percent of a portfolio and one representing thirty basis points enter with identical influence. Under this rule an investor is effectively the *set* of securities it holds, and a fund with one concentrated position and a long tail of small holdings is represented mostly by the tail, because the tail is where most of the tokens are. The portfolio shares that name the model—Portfolio-*Shares* BERT—determine only the order in which positions enter the sequence and which 62 survive truncation, not their influence on the resulting vector.

4.2 Weighted pooling

The natural generalisation replaces the uniform average with a weighted one,

$$x_i^w = \frac{\tilde{h}_i}{\|\tilde{h}_i\|}, \quad \tilde{h}_i = \sum_{\ell=1}^{L_i} w_{i\ell} h_{i\ell}, \quad w_{i\ell} \geq 0, \quad \sum_{\ell} w_{i\ell} = 1, \quad (2)$$

with $w_{i\ell}$ the share of position ℓ in investor i 's portfolio, taken directly from the sequence files of Section 3.3. Special tokens and padding receive weight zero, so that only holdings enter the average, while the attention mask continues to expose them to the encoder. Under this rule an investor is represented by what it has committed capital.

Relation to the source paper's open question. Gabaix et al. (2025) raise the magnitude question at the level of the *objective*, suggesting that log dollar holdings could replace ranked positions in what the model is trained to predict. Equation (2) intervenes later, at the level of *aggregation*, and is therefore a different treatment: the encoder still learns from ranks, and only the read-out is magnitude-aware. The compensating advantage is identification. Because both arms are estimated from the same sequences with the same encoder, any difference between the resulting partitions is attributable to aggregation alone, which a change of objective would confound with a change in what the encoder has learned. A magnitude-aware objective remains open for future research.

The two specifications differ *only* in the pooling operator: same holdings, same architecture, same seed, same training schedule, same clustering procedure, same context window, and in both cases the embedding is computed from the largest 62 positions. Wider coverage of the portfolio—aggregating all chunks, or enlarging the context window—is a separate question, deliberately held fixed here so that the comparison remains clean.

What weighting can and cannot change is worth stating before results are seen. Weighting redistributes emphasis *among securities an investor already holds*; it cannot introduce securities the investor was never able to hold. Where it should impact is in the relative influence of the head and the tail of a book: under equal weighting a broad index tracker is represented mostly by its several hundred small positions, under weighting by its mega-capitalisation core.

4.3 Model-based clustering on the sphere

4.3.1 Why a partition, and why model-based

Gabaix et al. (2025) inspect neighbourhoods: given a focal investor, who are its nearest neighbours? That is sufficient for illustration and insufficient for classification, which is the application this thesis takes up. A classification requires assigning every investor in the population to a group, which in turn requires a clustering model.

Unit-norm embeddings carry information only in their direction, so Euclidean methods are misspecified. Among spherical methods a *model-based* formulation is preferred here for three reasons. It returns *soft* assignments, acknowledging that some investors genuinely sit between groups. It attaches to each component a concentration parameter with a direct interpretation—how tightly a population huddles around its prototype—which turns out to be one of the more informative descriptors in Chapter 5 and has no counterpart in neighbourhood inspection. And it places the choice of the number of components inside likelihood inference.

4.3.2 The spherical Cauchy mixture

A random unit vector $x \in \mathbb{S}^{d-1}$ follows a spherical Cauchy distribution (Kato and McCullagh, 2020) with location $\mu \in \mathbb{S}^{d-1}$ and concentration $\rho \in [0, 1)$ if its density with respect to the uniform distribution on the sphere is

$$f(x \mid \rho, \mu) = \left(\frac{1 - \rho^2}{\|x - \rho\mu\|^2} \right)^{d-1}, \quad \|x - \rho\mu\|^2 = 1 + \rho^2 - 2\rho x^\top \mu. \quad (3)$$

The density is a monotone transformation of the cosine similarity $x^\top \mu$ —the same distance measure the fine-tuning stage of Section 4.1 was designed to validate, so the clustering model and the representation are consistent by construction. Investors aligned with the location direction are likely, misaligned ones unlikely, with ρ governing how steeply likelihood decays in the angle. At $\rho = 0$ the component is uniform on the sphere; as $\rho \rightarrow 1$ it concentrates toward a point mass at μ . Relative to the von Mises–Fisher family, the polynomial rather than exponential tail decay makes the model robust to outlying portfolios, and the absence of special-function normalising constants makes density evaluation cheap at scale (Sablica et al., 2025).

The K -component mixture is

$$f(x) = \sum_{j=1}^K \pi_j f(x \mid \rho_j, \mu_j), \quad \pi_j \geq 0, \sum_j \pi_j = 1, \quad (4)$$

estimated by the EM algorithm (Dempster et al., 1977): the E-step computes posterior component probabilities for each investor given current parameters; the M-step re-estimates (π_j, ρ_j, μ_j) from posterior-weighted data using the direct estimation algorithm of Kato and McCullagh (2020), as implemented in **circlus**’s `FLXMCspcauchy` driver. Estimation is embedded in **flexmix** (Leisch, 2004), whose `minprior` control eliminates components whose prior share falls below a threshold during iteration. The **flexmix** default of 5 % would forbid components smaller than roughly one thousand investors in these data; it is lowered to 1 % throughout.

The number of components is evaluated by the Bayesian information criterion (Schwarz, 1978) over a grid of K . Two caveats accompany BIC at this sample size and are examined

empirically in Section 5.3: the complexity penalty is small relative to attainable likelihood gains, and a mixture whose component shape only approximates the true within-group distribution may recruit additional components to absorb misfit rather than to represent additional populations.

4.4 Characterising the components

A partition is not an explanation. A mixture component is a set of investors whose embeddings point in similar directions; whether that similarity reflects a shared investment orientation, a shared regulatory wrapper, a shared geographic mandate or a shared parent company cannot be read off the geometry. This is the same problem Gabaix et al. (2025) solve with a language model, and the reason for solving it differently is that a narrative answer cannot be wrong in any usable sense: a prompt asking for what a set of funds has in common will always return something.

Every component is therefore described along the same four dimensions, using the attributes of Section 3.4, and the descriptions are then scored (Section 5.5.6). A label to each component is then given by feeding these attributes into a LLM which condenses them to summarize its characteristics.

Who the members are. Investor type, fund type, declared style and turnover classification. All are reported as *lift*: the share of a category within a component divided by its share in the population. Lift is used rather than raw shares because open-end funds are roughly three-quarters of the sample, so composition tables say “mostly OEF” everywhere and discriminate nothing. A lift of 2 means a category is twice as common inside the component as chance would predict.

How they invest. Portfolio breadth, assets under management, and the block of portfolio characteristics: beta, price-earnings and price-book ratios, dividend yield, sales growth, price momentum and relative strength. This block is the quantitative counterpart to the declared style label and is valuable precisely because it is independent of it: a component described as “Value” should, if the label means anything, also show low price-book and high dividend yield in its members’ measured characteristics. Agreement between a declared label and a measured profile is stronger evidence than either alone—and, as Section 5.7 shows, the two can come apart.

What they hold. Computed from the holdings joined to the security table: the country and region composition of positions, sector and industry lift, capitalisation-group mix, the median market capitalisation of holdings, and the issuers held by a disproportionate share of a component’s members, reported by company name. These descriptors matter because geography and capitalisation determine which securities can appear together in a portfolio

at all, and a model trained on co-holdings has no way to distinguish shared preference from shared eligibility. They are also the descriptors the global universe makes informative: in a US-only sample they would be nearly constant.

Who manages them. The managing institution of each fund, the share of a component accounted for by its three largest managers, the ultimate parent entity, and the declared investment mandate. A component dominated by one asset manager is a product lineup: sibling funds and share classes hold near-identical books for administrative reasons, and their embeddings are similar for reasons unrelated to investment behaviour. The declared mandate is a particularly clean descriptor, being a statement of the fund’s own constraints rather than an inference from what it happens to hold.

Labelling. The four blocks above, together with the fitted concentration ρ , produce for each cluster a page of tables and no name. A short label is obtained by passing exactly those tabulated quantities—the lift tables, the median characteristics, the country, region and capitalisation composition, the distinctive holdings and the manager concentration—to a large language model and asking for a phrase that describes them. Nothing else is supplied: no holdings text, no prospectus or strategy description, no fund or manager name beyond those appearing in the manager-concentration table, and no access to the embeddings themselves. The statistics are computed here; the model performs a condensation of them.

4.5 Ranking the organising dimensions

The descriptions above produce a large number of tables, and reading them is a matter of judgement. To make the central question answerable in a single comparison, every attribute is additionally scored by how strongly it explains component membership. This scoring is what distinguishes the procedure from a better-organised version of the source paper’s narrative approach: it allows competing explanations of the same component to be ordered.

For a categorical attribute the score is Cramér’s V computed on the attribute-by-component contingency table, with the bias correction of Bergsma (2013). Writing $\varphi^2 = \chi^2/n$ for the mean-square contingency of a table with r rows, c columns and n observations, the correction removes the expected value of φ^2 under independence and shrinks the table dimensions accordingly,

$$\tilde{V} = \sqrt{\frac{\tilde{\varphi}^2}{\min(\tilde{r}, \tilde{c}) - 1}}, \quad \tilde{\varphi}^2 = \max\left(0, \varphi^2 - \frac{(r-1)(c-1)}{n-1}\right), \quad (5)$$

with $\tilde{r} = r - (r-1)^2/(n-1)$ and $\tilde{c} = c - (c-1)^2/(n-1)$. The correction matters here: the uncorrected statistic rises mechanically with the number of categories, and the highest-cardinality attributes in this application are managing institution and ultimate parent—precisely the at-

tributes whose scores carry the most interpretive weight, since a high value would suggest the components are product lineups rather than investment groupings.

For a continuous attribute the score is the rank-based ϵ^2 : the share of total variance lying between components, computed on midranks rather than on levels. Writing R_i for the midrank of x_i in the pooled sample and $\bar{R}_j$ for the mean rank within component j ,

$$\epsilon^2 = \frac{\sum_j n_j (\bar{R}_j - \bar{R})^2}{\sum_i (R_i - \bar{R})^2}, \quad (6)$$

which is equivalent to $H/(n - 1)$ for the Kruskal–Wallis statistic H . The rank transform is not cosmetic. Computed on levels, the statistic is driven by squared deviations, and the heavy right tails of beta, price momentum and sales growth inflate the denominator enough to push the between-component share toward zero even where component medians differ plainly. Section 5.5.6 reports the attributes where this would have bound.

Both statistics lie in $[0, 1]$, with zero indicating independence from the partition and one indicating that the attribute determines it. Attributes measured at the holdings level are first aggregated to the investor—the share of an investor’s positions in US issuers, the share in its single largest sector, the median market capitalisation of its holdings—so that every score is computed on the same unit of observation. Because that aggregation can itself be done by position count or by portfolio weight, the holdings-level attributes are scored both ways and both figures are reported, so that the choice can be seen not to drive the ranking.

Coverage is reported alongside every score. Attributes drawn from the vendor’s classification tables are missing for part of the panel—assets under management most severely—and each score is computed on the investors for which the attribute is observed. A score resting on half the sample is not directly comparable to one resting on all of it, and the coverage column is what makes that visible. Where an attribute takes a single value across an entire sample, as the modal issuer country does in the US-only fit of Section 5.7, it is not scored at all.

Each attribute is then assigned to one of seven families: *breadth*, *size*, *geography*, *sector*, *artifact*, *vehicle* and *strategy*. The first four describe the investable universe an investor operates in—how many positions it carries, at what capitalisation, in which countries and which sectors. *Artifact* and *vehicle* describe administrative structure: the managing institution and ultimate parent in the first case, the fund and investor type in the second. *Strategy* is the largest family, with nine attributes, and describes investment orientation in the sense a practitioner would recognise—declared style, turnover regime, and the factor profile of the book. The families cut across the four description blocks of Section 4.4 instead of reproducing them, because the question they answer is different: not what can be said about a component, but what accounts for it. Each family is summarised by its strongest attribute, and the number of attributes over which that maximum is taken is reported, since a family with nine chances to lead is not on equal footing with one that has a single attribute. Comparing

families on that basis gives a legible answer to the question of which dimensions organise the geometry.

4.5.1 Limitations of the two measures

Four caveats govern how the ranking should be read.

First, $\tilde{V}$ and ϵ^2 are different statistics, so their values are not strictly comparable across the two kinds of attribute. The ordering within each kind is more reliable than a literal cross-kind comparison, and the family rollup—which mixes them—should be read as an ordering of magnitudes rather than a scale.

Second, the Bergsma correction removes the expected inflation of Cramér’s V under independence, but it does not make high- and low-cardinality attributes fully commensurable, and the corrected statistic can be sensitive in sparse tables where many cells are thinly populated. A leading score in the artifact family is therefore still treated as a flag prompting inspection of the manager concentration statistics reported alongside it, rather than as evidence in itself. Section 5.5.6 carries out that inspection.

Third, the rank transform buys robustness at the cost of information about magnitudes: ϵ^2 registers that components are ordered on an attribute, not how far apart they are. Two partitions that separate median market capitalisation by a factor of two and by a factor of fifty can receive similar scores. The component-level medians reported in the descriptive tables, not the score, are what convey the size of a difference.

Fourth, the maximum-within-family rollup is a deliberate simplification and is biased toward families with many attributes even after the count is reported, since the maximum of nine noisy scores exceeds the maximum of one in expectation. This works against the strategy family being ranked too low and in favour of breadth, which has a single attribute, being ranked too high—so the ranking reported in Chapter 5 is, if anything, conservative with respect to its own conclusion.

5 Clustering the investor embeddings

The methodology produced, for every reporting quarter, a matrix of investor embeddings on the unit sphere in $\mathbb{R}^{64}$. This chapter asks what those coordinates encode. The question is posed in three parts: whether the geometry contains group structure at all, how many groups it is reasonable to report, and what the groups turn out to be. The third question is the substantive one, and the answer will be that the components separate investors primarily by investable universe — geography and capitalisation band — rather than by investment orientation.

5.1 The reference quarter

All results in this chapter are computed on a single cross-section, 2025-Q4 (quarter end 31 December 2025). The choice is made for one reason: it is the terminal point of the annual dynamics panel analysed in Chapter 6, which runs from 2005 to 2025. Reporting the cross-section on the same quarter at which the dynamics chain ends means the components characterised here are the components whose histories that chapter traces, and no separate cross-section has to be introduced and reconciled.

The quarter is also the most fully populated in the panel. After the reference-table join, 99.9% of investors match an entry and 99.0% carry a declared style label.

Table 5 records the dimensions of the cross-section.

Table 5: The 2025-Q4 cross-section.

Investors with an embedding	32,389
Embedding dimension d	64
Positions in the top-62 slice	1,623,786
Issuers resolved to a name	100.0%
Investors matched in the reference table	99.9%
Investors with a declared style	99.0%

The universe is global. Table 6 reports how concentrated the panel is in US issuers, computed over positions whose issuer has an identified domicile.

Table 6: Investors by US share of the top-62 slice, 2025-Q4.

Threshold	Investors	Share of panel
$\geq 50\%$ US	11,227	34.7%
$\geq 75\%$ US	7,654	23.6%
$\geq 90\%$ US	4,984	15.4%
$\geq 95\%$ US	2,960	9.1%
$\geq 99\%$ US	880	2.7%

Only 15.4% of the panel is predominantly US by holdings. This is the single largest difference

between this study and Gabaix et al. (2025), whose universe is restricted to US equities by a CRSP merge applied before tokenisation. The consequences are taken up in Section 5.7.

5.2 Does the geometry contain structure?

Before fitting a mixture it is worth establishing that there is something to fit. A clustering algorithm returns components whether or not the data are grouped, and on a sphere in 64 dimensions the null case — points distributed uniformly — is not visually distinguishable from the alternative.

Attribute homogeneity of neighbourhoods. If the geometry encodes anything about investors, an investor’s nearest neighbours in cosine distance should share its observable attributes more often than chance. For each investor the share of its m nearest neighbours carrying the same declared style, the same domicile and the same fund type is compared to the panel base rate. At $m = 10$, the share of neighbours sharing declared style is 54.5%, against a base rate of 20.1% (lift $2.71\times$, coverage 99.0%); the same investor domicile is shared 57.6% of the time against a base rate of 12.2% (lift $4.72\times$, coverage 81.2%); and the same fund type 74.6% of the time against a base rate of 58.8% (lift $1.27\times$, coverage 97.7%). At $m = 50$ the lifts persist at a similar order — $2.38\times$, $4.34\times$ and $1.22\times$ respectively — declining only modestly as the neighbourhood widens. Domicile shows the sharpest lift of the three and fund type the weakest, though even fund type’s neighbours share type more often than chance by more than a fifth.

5.3 Selecting the number of components

5.3.1 Why the information criteria do not decide

A spherical Cauchy component in d dimensions costs approximately $d + 1$ parameters: $d - 1$ free coordinates for the mean direction on the sphere, one concentration parameter, and one mixing weight. The BIC penalty for an additional component is therefore about $(d + 1) \log n$. For computation reasons, the scan is fit on a fixed 5,000-investor subsample of the cross-section rather than on all $n = 32,389$ investors, so the relevant penalty uses $n = 5,000$: $(64 + 1) \log(5,000) \approx 554$.

The likelihood gains are of a different order entirely. On this subsample the median absolute BIC change per added component is 7,695 — a ratio of 13.9 times the penalty. BIC is therefore, to a good approximation, -2 times the maximised log-likelihood. It falls at every step across $k = 2, \dots, 20$ and never turns over. Any k reported as the BIC minimum would be the upper boundary of whichever grid happened to be searched.

The heavy tails of the Cauchy kernel compound the problem. No observation is ever badly misfit by a distant component, so no candidate k is penalised for leaving part of the sphere uncovered, and the likelihood improves smoothly with every additional degree of freedom.

5.3.2 The reported choice

Eight components are reported; the choice is made on interpretability.

One diagnostic bears on the choice without deciding it. The index-tracker component appears at $k = 4$ and $k = 8$ with essentially identical membership — 408 and 410 investors, both at $\rho \approx 0.97$, holding the same issuers at the same lifts. A component that survives a doubling of the resolution unchanged is not an artefact of either. At $k = 12$ it is still recovered, at $\rho = 0.977$ with the same distinctive holdings, but with 356 members, about 13% fewer. The core of the component is stable across all three resolutions; its boundary is not.

Twelve components resolve genuine additional structure — India, an emerging-markets financial group, and a cleaner split of the US components — but they do so at a cost the eight-component solution avoids. Two of the twelve components are mixtures of unrelated investor groups rather than coherent segments (Section 5.6), one component carries no distinctive holdings at all beyond the residual core, and the index-tracker component’s core, though recovered at the same ρ , shrinks by roughly 13% as members are drawn into neighbouring components close to the `minprior` floor. Eight components is reported as the specification that resolves the geography and capitalisation axes documented above without yet fragmenting into partially-mixed groups; the twelve-component fit is reported alongside, in Section 5.6, as evidence of what finer resolution adds and where it stops being clean.

5.4 Baseline and weighted pooling compared

Two sets of embeddings are available for every quarter. The baseline arm applies unweighted mean pooling over the token representations, replicating Gabaix et al. (2025). The weighted arm, the extension developed in Section 4.2, weights each position by its share of the portfolio. Both are produced by the same training script and differ in the pooling flag alone, so a comparison isolates the effect of pooling.

Table 7 reports the baseline arm fitted at $k = 8$ on the global universe at 2025-Q4 ($N = 32,389$ investors). Both restarts retained all eight components. The solution consists of one tight component and seven diffuse ones. Component 1 contains 412 investors and has $\rho = 0.966$. Three quarters of its members carry an index style label, their median portfolio has 500 positions, effectively all of their positions are mega-caps, 97.6% are in US issuers, and its most distinctive holdings (McDonald’s, AT&T, Amgen) are each held by more than 90% of members. These are US large-cap index trackers, whose near-identical portfolios map to near-identical points on the sphere. Component 1 is also the only component within 1.5 times the 324-investor threshold below which `minprior` = 0.01 removes a component, and it is therefore the one most exposed to being absorbed under other initialisations. The remaining seven components are of similar size (3,582 to 5,406 investors) and have ρ between 0.236 and 0.378. The most diffuse, component 7, is a residual group of narrow portfolios (median 28 positions) without distinctive holdings: the names its members hold

Table 7: Baseline arm, global universe, $k = 8$ (2025-Q4)

<i>Panel A: Components</i>						
Component	n	Share (%)	ρ	Median positions	Truncated (%)	Median cap (USD bn)
1	412	1.3	0.966	500	100.0	292.6
2	4,925	15.2	0.337	51	30.6	4.9
3	5,406	16.7	0.279	57	42.1	11.2
4	3,582	11.1	0.378	131	94.4	38.1
5	4,090	12.6	0.359	95.5	88.0	74.9
6	4,331	13.4	0.301	106	86.5	22.2
7	5,252	16.2	0.236	28	0.2	16.5
8	4,391	13.6	0.377	45	0.1	59.1
All	32,389	100.0				

<i>Panel B: Association ranking rolled up by family</i>				
Family	Strongest attribute	Statistic	Strength	Scored
Breadth	Number of positions	ε^2	0.613	1
Artifact	Managing institution	V	0.368	2
Geography	Modal issuer country	V	0.314	4
Size	Median holding market cap	ε^2	0.276	2
Strategy	Declared style	V	0.180	9
Sector	Declared specialisation	V	0.172	2
Vehicle	Fund type	V	0.139	3

Notes: Spherical Cauchy mixture fitted to the mean-pooled (baseline) investor embeddings at 2025-Q4, $N = 32,389$; both restarts retained eight components. ρ is the component concentration. Median positions counts the full portfolio; Truncated is the share of members whose portfolio exceeds the 62-position slice read by the encoder; Median cap is the median market capitalisation of the positions held by component members. Strength is rank ε^2 for continuous attributes and bias-corrected Cramér's V for categorical ones; each family reports its strongest attribute, and Scored gives the number of attributes in the family.

most often are global mega-caps, which they hold less often than the panel as a whole.

The family rollup in Panel B places breadth first by a wide margin. The number of positions alone reaches $\varepsilon^2 = 0.613$, even though it is the only attribute in its family. Manager identity follows ($V = 0.368$), then modal issuer country (0.314) and median holding capitalisation (0.276). The manager score does not indicate fund-family silos. The three largest managers never account for more than 11.7% of a component, or more than 6.4% outside component 1, so the score reflects sibling funds of the same manager falling into the same component. The geographic axis is visible in the components themselves. Component 4 is 39.9% China-domiciled. Component 8 over-represents French, German and Spanish funds holding European large caps, and component 6 over-represents funds domiciled in India, Hong Kong and South Korea. Component 2 holds the smallest companies (median USD 4.9 billion) and over-weights Thai, Israeli and Brazilian issuers. Strategy ranks only fifth. Declared style reaches 0.180 and beta, the strongest measured characteristic, reaches 0.159, even though this family has nine attributes over which the maximum is taken.

A ranking led by breadth identifies neither segmentation nor strategy groups. The baseline solution is best described as organised by portfolio width, with fund family, geography and capitalisation band as secondary axes. Part of the breadth result, however, is likely to reflect input construction. The encoder reads at most 62 positions per investor, and the components divide sharply on whether that cap binds. Between 86.5% and 100% of members are truncated in components 1, 4, 5 and 6, against 0.1% and 0.2% in components 7 and 8, with components 2 and 3 in between. A portfolio's width is therefore visible to the encoder as the length of its input sequence, independently of the securities it contains. The breadth association should be read as a property of the baseline representation before it is read as a property of investor behaviour.

5.5 The eight components of the weighted global arm

This section is the substantive core of the chapter. The mixture is fitted on all 32,389 investors of the 2025-Q4 cross-section using the weighted embeddings, at $k = 8$, retaining all eight components across restarts. Table 8 reports sizes and concentrations.

The concentration parameters divide the components into two regimes. Component 5 sits at $\rho = 0.970$, some twenty times tighter on the $1 - \rho$ scale than anything else; the remaining seven fall between 0.238 and 0.409. This reflects a genuine feature of the data: index trackers hold near-identical portfolios, and everything else is a slice of a continuum.

5.5.1 Component 5: US mega-cap index trackers

The smallest and by far the tightest component. It is 74.4% Index by declared style (lift 3.14) and 95.6% Very Low turnover, and its positions are 99.2% US issuers and 99.9% mega-capitalisation, with a median holding market capitalisation of \$292.6bn. Median book length

Table 8: Component sizes and fitted concentration, weighted arm, global universe, 2025-Q4, $k = 8$.

Component	Label	Size	Share	ρ
1	China mid-cap growth	4,258	13.1%	0.386
2	US mega-cap growth	3,018	9.3%	0.409
3	US income and real assets	4,956	15.3%	0.290
4	Asia–Europe value	5,396	16.7%	0.287
5	US mega-cap index trackers	410	1.3%	0.970
6	Asian technology	4,498	13.9%	0.274
7	Concentrated global core	5,625	17.4%	0.238
8	European quality growth	4,228	13.1%	0.367

is 500 names and every member is truncated by the 62-token context window.

Its distinctive holdings are held by 58–98% of members at lifts of 16.4 to 18.5 — McDonald’s, AT&T, Amgen, GE Vernova, Verizon, Raytheon, Philip Morris, Boeing. No other component approaches this. Exchange-traded funds are overrepresented at a lift of 1.98 and variable annuity funds at 4.25; the largest single manager is SSgA at 4.3% and the top three managers account for 11.8%, the highest administrative concentration in the panel. The declared mandate region carries a lift of 11.6.

Every lens agrees, and the component is recovered identically at $k = 4$. It is the clearest evidence in the chapter that the embeddings encode portfolio content rather than noise.

5.5.2 Components 1, 6: Asian equity, split by capitalisation

Component 1 (4,258 investors) is Chinese mid-cap growth. Its positions are 59.0% Chinese issuers at a lift of 2.67 and 60.6% Asian by region, with 35.4% in mid-cap and only 17.8% in mega-cap names; the median holding is \$6.3bn, the smallest of any component. Its distinctive holdings are domestic Chinese industrials and materials — Sinomine Resource, EVE Energy, Zhejiang Huayou Cobalt, XCMG Construction Machinery — at lifts of 4.1 to 5.6. It is 40.5% Aggressive Growth by declared style (lift 1.75) and carries the highest turnover in the panel, with High turnover at a lift of 2.43. Investor domicile is 57.4% Chinese, and the largest manager is E Fund Management.

Component 6 (4,498) is Asian technology at a higher capitalisation band: 50.0% Asian positions but 62.4% mega-cap and a median holding of \$35.5bn, five times component 1. Semiconductors carry a lift of 1.91 at 15.5% of positions, and the distinctive holdings are the regional technology majors — MediaTek, Hon Hai, Xiaomi, PDD, NetEase, Delta Electronics — at lifts of 4.2 to 5.2. Declared mandate region carries a lift of 4.74.

The two components share a region and separate on capitalisation and domicile. This is the pattern that recurs throughout the solution.

5.5.3 Components 2, 3: US equity, split by orientation

Component 2 (3,018) is US mega-cap growth: 69.2% North American positions, 84.5% mega-cap, and a median holding of \$172.6bn, the highest outside the index component. It is 55.6% Growth by declared style at a lift of 2.09 — the strongest style association among the diffuse components — with the highest non-index beta at 0.92, price momentum at 0.76 and price-to-book at 6.51. Its distinctive holdings are US mega-caps held by a large fraction of members: Eli Lilly by 69%, JPMorgan by 68%, Tesla by 53%, AbbVie by 50%, at lifts around 5. Median book length is 168 names and 94.4% of members are truncated.

Component 3 (4,956) is the income and real-assets counterpart: 53.0% North American positions but a median holding of \$21.1bn, and a sector profile dominated by real estate investment trusts at 10.1% of positions and a lift of 4.75, followed by oil and gas pipelines (3.16) and railroads (2.80). It is 19.9% Yield by declared style at a lift of 2.55, the strongest single style lift among the diffuse components, with Deep Value at 1.84 and Value at 1.59. Beta is 0.51 and dividend yield the highest in the panel at 0.77; turnover is 69.2% Very Low.

Components 2 and 3 are the clearest case in the solution of two components that share a geography and separate on something closer to investment orientation. Their distinctive-holdings tables differ in kind: component 2's are held by half to two thirds of its members at high lift, while component 3's are largely generic US large-caps at lifts near or below 1, with only Prologis at 3.0. The identity of component 3 rests on its sector and style profile rather than on specific names.

5.5.4 Components 4, 8: Europe and the developed cross-region

Component 4 (5,396) is Asia–Europe value: 52.0% Asian and 26.3% European positions, the most balanced capitalisation profile in the panel (28.7% mega, 28.4% mid, 24.2% large, 17.1% small) and a median holding of \$6.6bn. It has the lowest beta (0.32), the lowest price momentum (0.42) and the lowest sales growth (5.28) of any component. It is 18.2% GARP by declared style at a lift of 2.08. Its distinctive holdings are European financials and defensives — BBVA, Intesa Sanpaolo, UniCredit, BNP Paribas, ING, Munich Re, GSK — at lifts of 1.9 to 2.2.

Component 8 (4,228) is European quality growth: 35.9% European positions, the highest in the panel, with 52.2% mega-cap and a median holding of \$38.2bn. It is 38.2% Growth at a lift of 1.44, and its distinctive holdings are the European quality compounders — Novo Nordisk, L'Oréal, Schneider Electric, RELX, Infineon, Saint-Gobain — at lifts of 2.2 to 2.7. Investor domicile is 23.2% Luxembourg, with France at a lift of 2.76 and Spain at 2.24, which reflects the UCITS wrapper rather than the location of the managers themselves.

5.5.5 Component 7: the concentrated global core

Component 7 (5,625) is the largest and the least distinctive, and it is reported as such. Its regional profile is the most balanced in the panel (36.8% North America, 34.7% Asia, 19.6% Europe), and every one of its top holdings carries a lift *below* 1: Alphabet at 0.9, Microsoft at 0.8, NVIDIA at 0.7, Apple at 0.6. It holds the global mega-caps less often than the panel average.

What distinguishes it is not what it holds but how much of it. Median book length is 29 names, the narrowest in the panel, and only 0.3% of members exceed the context window. Median assets under management are the lowest of any component, and administrative concentration is the lowest as well, with the top three managers accounting for 2.8% across 1,483 distinct managers. Hedge funds are overrepresented at a lift of 2.07 and Very High turnover at 1.62.

The component is best described as small, concentrated, globally diversified funds holding the modal international portfolio. It is a residual in the sense that it has no distinctive holdings, but it is not a failure of the fit: a component consisting of investors who hold the consensus portfolio in concentrated form is a substantively meaningful group.

5.5.6 What explains component membership

Each observable attribute is scored for how strongly it explains component membership. Categorical attributes use Cramér's V with the bias correction of Bergsma (2013), which is necessary here because the artifact family — managing institution, ultimate parent — is precisely the high-cardinality one, and the uncorrected statistic rewards cardinality as such. Continuous attributes use rank-based ϵ^2 rather than η^2 , because a small number of extreme values in beta and momentum otherwise dominate the total sum of squares and drive the between-component share to zero.

Rolled up by family, the ordering is breadth (0.521), artifact (0.386) and size (0.386), geography (0.316), strategy (0.201), sector (0.192) and vehicle (0.140). The answer to the question that motivates the ranking is therefore neither “strategy” nor cleanly “segmentation”: the components are organised first by portfolio width and capitalisation band, which is to say by investable universe.

Two qualifications are needed.

The first concerns portfolio breadth, which is the strongest single attribute and also the one most entangled with the encoder itself. The share of each component truncated by the 62-token context window ranges from 0.3% to 100%, and tracks the components closely. It is therefore not currently possible to separate the hypothesis that diversified investors hold systematically different portfolios from the hypothesis that the encoder learned less about investors whose books it truncated.

Table 9: Attributes ranked by association with component membership. Weighted arm, global universe, 2025-Q4, $k = 8$. Categorical attributes scored by bias-corrected Cramér's V , continuous by rank ϵ^2 .

Attribute	Kind	Strength	Coverage	Family
n_assets_full	continuous	0.521	1.000	breadth
mgr_entity_proper_name	categorical	0.386	0.957	artifact
med_cap	continuous	0.386	0.995	size
modal_issuer_country	categorical	0.316	1.000	geography
inv_country_desc	categorical	0.293	0.812	geography
fs_ultimate_parent	categorical	0.288	0.813	artifact
inv_obj_region_code	categorical	0.210	0.818	geography
style_any	categorical	0.201	0.990	strategy
inv_obj_spec_code	categorical	0.192	0.877	sector
pct_us	continuous	0.164	1.000	geography
beta	continuous	0.159	0.933	strategy
turnover_any	categorical	0.150	0.999	strategy
fund_type_desc	categorical	0.140	0.977	vehicle
pb_ratio	continuous	0.140	0.931	strategy
investor_type	categorical	0.127	1.000	vehicle
sales_growth	continuous	0.126	0.934	strategy
price_momentum	continuous	0.120	0.934	strategy
relative_strength	continuous	0.104	0.925	strategy
aum_any	continuous	0.092	0.456	size
pct_top_sec	continuous	0.064	1.000	sector
pe_ratio	continuous	0.036	0.932	strategy
dividend_yield	continuous	0.035	0.934	strategy

The second concerns the artifact family. Managing institution scores 0.386, second only to breadth. This does not indicate that the components are product lineups: the largest single manager holds between 1.0% and 4.3% of any component, and the top three managers account for between 2.8% and 11.8%. The high score reflects the fact that a manager's funds share an investable universe and a house allocation, which is exactly the structure the embeddings are designed to detect, rather than a mechanical grouping of one firm's products.

Finally, the sector concentration measure `pct_top_sec` scores 0.064 portfolio-weighted against 0.067 count-weighted, and the capitalisation measure 0.386 against 0.375. The weighting choice does not drive any conclusion in this table.

5.6 A finer partition: twelve components

At $k = 12$ the full cross-section is fitted with all twelve components retained in both restarts (Table 10). The smallest component has 356 members against a `minprior` floor of 324, within a factor of 1.5, so the floor constraint is close to binding.

Table 10: Component sizes and fitted concentration, weighted arm, global universe, 2025-Q4, $k = 12$. Labels are provisional.

Component	Label	Size	Share	ρ
1	European quality growth	3,415	10.5%	0.392
2	Developed large-cap growth	2,213	6.8%	0.460
3	US mega-cap core	986	3.0%	0.575
4	Indian domestic equity	1,769	5.5%	0.433
5	US mega-cap index trackers	356	1.1%	0.977
6	Concentrated global core	5,049	15.6%	0.248
7	US value and insurance	2,559	7.9%	0.355
8	Growth technology and hedge funds	3,774	11.7%	0.285
9	European financials and fund-of-funds	3,343	10.3%	0.311
10	China mid-cap growth	3,405	10.5%	0.421
11	Emerging-markets financials	1,759	5.4%	0.440
12	Developed mid-cap income	3,761	11.6%	0.332

Four of the eight components have recognisable successors on their profiles. The index trackers (component 5) are recovered at $\rho = 0.977$, holding the same issuers at lifts of 16.7 to 19.5, but with 356 members rather than 410. European quality growth reappears as component 1, with the same distinctive holdings — Novo Nordisk, L'Oréal, Schneider Electric, RELX, Infineon, Saint-Gobain — at lifts of 2.3 to 2.8 and the same Luxembourg and French domiciles. The concentrated global core reappears as component 6. It is again the largest component, with a median book of 28 names, the lowest assets under management, and top holdings at lifts below 1. Chinese mid-cap growth reappears as component 10: 67.3% Chinese positions, the smallest median holding at \$5.9bn, domestic materials and industrials at lifts of 5.8 to 6.9, and the highest turnover. It is now joined by Israeli and

Brazilian domestic funds, at domicile lifts of 4.82 and 3.54.

Three of the new components are sharply defined. Component 4 is Indian domestic equity: 57.5% of positions in Indian issuers at a lift of 11.2, a further 12.7% in Japan, 45.8% Indian domicile at a lift of 13.4, GARP at a lift of 4.38, and distinctive Indian mid-caps — Glenmark, Fortis Healthcare, Multi Commodity Exchange, CG Power — at lifts of 15.3 to 17.5. These investors appear to have sat in the $k = 8$ Asia–Europe value component, whose GARP lift and low beta they would account for; the European financials that defined that component’s holdings reappear in component 9. Component 11 is an emerging-markets component: 83.0% Asian positions, 44.6% Index, declared mandate region R at a lift of 12.1, and distinctive holdings in Gulf and Taiwanese financials — Qatar National Bank, Al Rajhi, Saudi National Bank, Fubon, Cathay, CTBC — at lifts of 15.3 to 16.5. Component 3 is a US mega-cap core component sitting between the index trackers and the growth funds: 93.1% North American positions, 97.8% mega-cap, 51.2% Growth and 42.0% Index, at $\rho = 0.575$. Its distinctive holdings — Home Depot, Morgan Stanley, Goldman Sachs, IBM, Caterpillar — are held by 57% to 75% of members at lifts near 10. With component 3, the two-regime description of concentration no longer holds at this resolution.

The $k = 8$ US mega-cap growth component appears to divide between component 3 and component 2. Component 2 is developed-market large-cap growth: 55.6% North American and 34.9% European positions, 88.5% global mandates, and European domiciles, with Luxembourg at 24.3% and the Netherlands, Denmark and Austria at lifts of 3.6 to 3.9. Its distinctive holdings mix US and European large-caps — JPMorgan, Roche, Novartis, NatWest, BAE Systems — at lifts of 4.0 to 4.5. The US income component appears to divide into two parts. The first is component 7, US value: 83.1% North American positions, Value and Yield at lifts of 2.18 and 1.73, variable annuity funds at 3.49, and US insurers and financials — AIG, Allstate, Hartford, Synchrony — at lifts of 6.5 to 7.7. The second is component 12, developed mid-cap income: 41.4% European positions, real estate investment trusts at 8.3% (lift 3.91), Yield at a lift of 2.22, Nordic domiciles, and the lowest momentum and relative strength in the panel. Component 12 has no issuer held by 10% or more of its members.

Two components are mixtures. Component 8 combines US growth software — Snowflake, CrowdStrike, AppLovin, Synopsys — with a Thai domestic fund market. Thai issuers make up 9.3% of its positions (lift 6.17), 9.5% of its members are Thai-domiciled (lift 6.34), and it has the highest hedge-fund concentration in the panel (lift 4.29). Component 9 combines European financials — UniCredit, Munich Re, Deutsche Bank — with domestic Chinese industrials and a fund-of-funds exposure, investment trusts and mutual funds accounting for 10.9% of positions at a lift of 3.82. The $k = 8$ Asian technology component has no evident successor: none of its distinctive holdings appears among the leading distinctive holdings of any $k = 12$ component.

Relative to $k = 8$, twelve components add three sharply defined groups and split the two diffuse US groups into more coherent parts. They also produce two heterogeneous com-

ponents, a second component without distinctive holdings, and an index-tracker component close to the floor.

5.7 Restricting the universe to US-focused investors

The results above are computed on a global universe, which is the principal departure of this study from Gabaix et al. (2025). That paper restricts to US equities by merging with CRSP before tokenisation, so non-US securities are never seen by the encoder. That design cannot be reproduced here without regenerating the sequences and retraining.

What can be done is a weaker, post-hoc analogue: select investors whose top-62 slice is at least 90% US issuers — 4,984 investors, 15.4% of the panel — and refit the mixture on that subset alone. Geographic variation is thereby removed from the *clustering*, though it remains present in the *representation*, since the encoder was trained on the global universe. The question the restriction answers is: once geography is held roughly fixed, what organises the remaining investors?

Restricting the sample to investors with at least 90% of their positions in US issuers retains 4,984 of the 32,389 investors (15.4%), with a median US share of 0.952, and 262,893 positions. The weighted arm is refitted on this subsample at $k = 8$. Because the sample is smaller, `minprior` is raised from 0.01 to 0.03 so that the smallest admissible component still holds 150 investors. Both restarts retained all eight components. Table 11 reports the result.

Two components are tight and are defined by vehicle rather than by orientation. Component 4 (368 investors, $\rho = 0.976$) consists of US large-cap index trackers. 78.0% of its members carry an index style label, the median portfolio has 500 positions, and its most distinctive holdings are each held by 89% to 99% of members. The same group appears in the global solutions. Component 7 (165, $\rho = 0.877$) is administrative. Every member is an open-end fund managed by Fidelity Management & Research, and all of its positions are in mutual funds, predominantly Fidelity Series funds. These funds of funds pass the 90% filter because the funds they hold are US-registered, although the underlying exposure includes international equity, bonds and commodities. At 165 investors, component 7 is also the only component close to the removal threshold.

The remaining six components separate by capitalisation band first and by orientation second, with ρ between 0.261 and 0.607. Two of them hold mega-caps. Component 5 (596) is broad mega-cap growth: 97.3% of its positions are mega-caps, the median portfolio has 252 positions, and 55.2% of members are labelled Growth (lift 1.76). Component 6 (765) is concentrated technology growth. Its median portfolio has 58 positions and Aggressive Growth is over-represented at a lift of 3.19. It has the highest median beta (1.23), price-to-book (8.69) and sales growth (16.8) in the panel, and its distinctive holdings are software names such as Snowflake, CrowdStrike, Cadence and Synopsys. Its 12.1% Luxembourg domicile (lift 2.39) again reflects the UCITS wrapper. Component 3 (728) is large-cap value and income. Value

Table 11: US-only fit. Weighted arm, investors with at least 90% of positions in US issuers, 2025-Q4, $k = 8$.

<i>Panel A: Components</i>							
Comp.	n	Share (%)	ρ	Median positions	Truncated (%)	Median cap (\$bn)	Description
1	1,430	28.7	0.261	37	11.6	25.4	Concentrated residual
2	590	11.8	0.551	110	90.3	5.2	Mid-cap, value tilt
3	728	14.6	0.488	111	97.1	48.8	Large-cap value and income
4	368	7.4	0.976	500	100.0	292.6	Large-cap index trackers
5	596	12.0	0.602	252	96.3	215.0	Mega-cap growth
6	765	15.3	0.403	58	45.1	123.2	Technology growth
7	165	3.3	0.877	31	0.0	—	Fidelity funds of funds
8	342	6.9	0.607	398	98.0	14.9	Mid-cap, index-heavy
All	4,984	100.0					

<i>Panel B: Association ranking rolled up by family</i>					
Family	Leading attribute	Statistic	Strength	Scored	
Size	med_cap	ϵ^2	0.743	2	
Breadth	n_assets_full	ϵ^2	0.608	1	
Sector	inv_obj_spec_code	V	0.422	2	
Artifact	mgr_entity_proper_name	V	0.376	2	
Strategy	sales_growth	ϵ^2	0.364	9	
Vehicle	inv_obj_asset_type_code	V	0.261	3	
Geography	pct_us	ϵ^2	0.144	3	

Notes: $N = 4,984$; minprior = 0.03; both restarts retained eight components. Median positions counts the full portfolio; Truncated is the share of members whose portfolio exceeds the 62-position slice; Median cap is the median market capitalisation of the positions held by component members. Component 7 holds only fund shares, which carry no market capitalisation or portfolio characteristics, so it does not enter the size and strategy scores. Strength is rank ϵ^2 for continuous attributes and bias-corrected Cramér's V for categorical ones. modal_issuer_country takes a single value in this sample and is not scored.

and Yield are over-represented at lifts of 2.18 and 1.95, it has the highest median dividend yield (0.90), and its distinctive holdings include Archer-Daniels-Midland, Consolidated Edison, General Mills and CF Industries. Further down the capitalisation range, component 8 (342) holds a median company of \$14.9bn in portfolios of median 398 positions, and 51.2% of its members are labelled Index. Component 2 (590) holds the smallest companies (median \$5.2bn, 60.7% of positions mid-cap), tilts towards Value (lift 2.20), and has distinctive holdings at lifts of 7 to 8. Component 1 (1,430, $\rho = 0.261$) is the largest and most diffuse, and it mirrors the concentrated residual of the global solution. Its median portfolio has 37 positions, hedge funds are over-represented at a lift of 2.54 and High turnover at 2.08, and it has no distinctive holdings: the names its members hold most often are mega-caps held at lifts below 1.

Panel B places size first (`med_cap`, 0.743) and breadth second (0.608). Declared specialisation (0.422) and manager identity (0.376) follow, then the strategy family, led by sales growth (0.364) with price momentum close behind (0.340). Vehicle (0.261) and geography (0.144) close the ranking. The holdings-based geography attributes have almost no variance left by construction, so their scores are not comparable to the global ones. Size and breadth lead again, but capitalisation means something different here. Across national markets it proxies the investable universe; within a single market it is the size axis of the style box, so part of the leading score is itself orientation.

Two features of the restricted run should be noted whichever way the results fall. The holdings-based geography measures lose their variance by construction: `modal_issuer_country` becomes constant and `pct_us` is bounded below at 0.90, so their scores are not comparable to the unrestricted run and only the ordering of the remaining families should be read across the two. Investor domicile, by contrast, retains variation, because non-US managers run US mandates — the Irish and Luxembourg domiciles that appear in the unrestricted solution reflect UCITS wrappers rather than the location of the investment decision.

6 Cluster Dynamics Across Quarters

Section 5.5 answers the first half of the research question for one cross-section: the components are economically distinct, and the attributes that distinguish them can be ranked. This chapter takes up the second half. A partition of a single quarter is compatible with two different pictures. The components may be durable populations that persist while individual investors occasionally move between them, in which case the classification describes something real about the structure of the market. Or they may be a summary of one quarter's geometry, reassembled differently every time the encoder is retrained, in which case the descriptions of Section 5.5 apply to 2025-Q4 and no further. Distinguishing the two requires tracking investors across the panel, and the question of interest is not only whether membership persists on average but when it does not: whether movement concentrates in identifiable periods rather than arriving at a constant rate.

The source framework does not pose this question. Gabaix et al. (2025) estimate one model per cross-section and interpret the embeddings within a quarter. The per-quarter design is inherited here, and it is precisely what makes the dynamic question non-trivial: two independently trained encoders produce coordinate systems that have nothing in common. The tool used to overcome this is borrowed, once more, from natural language processing. Hamilton et al. (2016) train a separate word-embedding model on each decade of a historical corpus, align consecutive decades by an orthogonal rotation, and measure how far each word moves between them. This chapter transfers that design to investors. It makes one substantive change, in the choice of the objects the rotation is estimated from, and adds one validation that has no counterpart in the linguistic setting: investors, unlike words, choose their own contexts, and the holdings they trade can be observed directly.

The analysis uses the weighted arm of Section 4.2, the global universe, and the fourth quarter of every year from 2005 to 2025. Comparing fourth quarters only keeps the semi-annual reporters of Section 3.5 in both end points of every comparison, so that the composition of each step is not distorted by the disclosure calendar. The panel therefore consists of 21 cross-sections and 20 consecutive year pairs.

6.1 The alignment problem

Each PS-BERT model is trained from a fresh initialisation. The 64 coordinates of the 2024 model and the 64 coordinates of the 2025 model therefore have no common meaning, and an investor's two vectors cannot be compared directly even if its portfolio did not change. What the two spaces can share is their internal arrangement. The contrastive objective of Section 4.1.1 makes cosine similarity the operative notion of distance, and a map that preserves all cosines is an orthogonal transformation. If the two models have learned the same arrangement of investors, the spaces differ by a rotation, possibly combined with a reflection.

Let $B \in \mathbb{R}^{m \times 64}$ hold the unit-norm embeddings in year t of a set of m *anchor* investors, and A the embeddings of the same investors in year $t + 1$, row for row. The orthogonal Procrustes problem (Schönemann, 1966) is

$$R_{t \rightarrow t+1} = \arg \min_{R^\top R = I} \|BR - A\|_F, \quad B^\top A = UDV^\top \Rightarrow R_{t \rightarrow t+1} = UV^\top, \quad (7)$$

and $x_i(t) R_{t \rightarrow t+1}$ expresses investor i 's year- t embedding in the coordinates of year $t + 1$. Because all rows have unit length, minimising the Frobenius distance is the same as maximising the summed cosine between each anchor's rotated year- t vector and its year- $t+1$ vector, so the alignment optimises exactly the similarity the encoder was trained to make meaningful. Reflections are admitted, as in Hamilton et al. (2016). A rotation also maps the unit sphere onto itself, which has a convenient consequence for the mixture model: the spherical Cauchy parameters of Section 4.3 transform as $\mu \mapsto \mu R$ with ρ unchanged, so the Chapter 5 model can be evaluated on any aligned year without re-estimation.

Rotations are estimated between consecutive years only. A year t is carried into the 2025 frame by the product of the consecutive rotations,

$$R_{t \rightarrow 2025} = R_{t \rightarrow t+1} R_{t+1 \rightarrow t+2} \cdots R_{2024 \rightarrow 2025}, \quad (8)$$

so no pair of distant years ever needs anchors of its own. The price is that errors in the early steps accumulate along the chain.

The one decision the method leaves open is which investors serve as anchors. Least squares attributes every difference between an anchor's two vectors to the rotation, so an anchor that genuinely moved pulls the rotation in the wrong direction. Hamilton et al. (2016) align on the whole shared vocabulary and rely on most words not changing meaning between adjacent decades. Investors offer a better option, because whether an investor "changed" can be measured from its holdings, independently of the embeddings. The rest of this section builds on that observation.

6.2 Holdings stability and the choice of anchors

6.2.1 Measuring stability

For an investor present in both years, let $w_{ia}(t)$ be the weight of issuer a in its top-62 slice, renormalised to sum to one within the slice as in the weighted pooling of Section 4.2. Holdings stability is the weighted overlap

$$\omega_i(t) = \sum_a \min(w_{ia}(t), w_{ia}(t+1)) = 1 - \frac{1}{2} \sum_a |w_{ia}(t+1) - w_{ia}(t)|, \quad (9)$$

which equals one minus the share of the slice that would have to be traded to turn the year- t weights into the year- $t + 1$ weights. Turnover is defined as $\tau_i(t) = 1 - \omega_i(t)$. The

measure is chosen to match what the weighted embedding actually encodes: weighted pooling responds to money moving between names, not to the rank order of two near-equal positions changing.

The measure has a ceiling that matters for everything that follows. A fund that never trades still sees its weights drift, because its holdings earn different returns. For a buy-and-hold slice, $w_{ia}(t+1) = w_{ia}(t)(1+r_a)/(1+\bar{r}_i)$, so $\omega_i(t) = 1 - \frac{1}{2} \sum_a w_{ia}(t) |r_a - \bar{r}_i| / (1 + \bar{r}_i)$, which with the typical cross-sectional dispersion of one-year returns lies in the high 0.8s to low 0.9s. Positions that cross the rank-62 boundary lower it further. A perfectly passive fund therefore does not score one. In the 2024–2025 pair, the 156 investors whose top-62 name set is literally unchanged have a median ω of 0.926. Values clearly above 0.90 require something that resets the weights: equal-weight products rebalanced to equal shares, or funds of funds managed to a target allocation.

6.2.2 Who is stable

Table 12 describes ω for the 28,709 investors present in both 2024-Q4 and 2025-Q4. The median is 0.645, the quartiles are 0.430 and 0.784, and 149 investors (0.5%) share no position at all between the two quarters. The latter are mostly open-end funds with unchanged portfolio size and, in several cases, an index mandate, and are more plausibly identity breaks in the source data than genuine trading.

Table 12: Holdings stability ω between 2024-Q4 and 2025-Q4 by declared turnover class, declared style and Chapter 5 component. “Stable” is $\omega \geq 0.85$. Groups with fewer than 50 investors are omitted.

	Group	<i>n</i>	Median ω	Share stable
Turnover	Very Low	16,141	0.749	0.207
	Low	5,010	0.618	0.033
	Medium	4,627	0.419	0.010
	High	2,627	0.197	0.006
	Very High	281	0.273	0.025
Style	Index	6,597	0.799	0.322
	Yield	2,186	0.697	0.113
	GARP	2,538	0.655	0.060
	Growth	7,565	0.629	0.090
	Deep Value	488	0.605	0.061
	Value	2,414	0.590	0.049
	Aggressive Growth	6,663	0.492	0.033
Component	5 US mega-cap index trackers	361	0.880	0.920
	6 Asian technology	4,003	0.730	0.191
	3 US income and real assets	4,487	0.695	0.175
	2 US mega-cap growth	2,628	0.686	0.100
	4 Asia–Europe value	4,842	0.667	0.123
	8 European quality growth	3,779	0.637	0.053
	7 Concentrated global core	4,635	0.594	0.109
	1 China mid-cap growth	3,974	0.360	0.033

The measure agrees with three descriptions of the funds that never enter its computation. Median stability falls monotonically across FactSet’s four large turnover classes, from 0.749 for Very Low to 0.197 for High. Index funds are the most stable style and Aggressive Growth the least. Among the Chapter 5 components, the US index trackers are the most stable (0.880) and the China mid-cap growth component, which Section 5.5 describes as the highest-turnover group, the least (0.360). The ordering supports reading ω as a measure of genuine portfolio persistence.

The same table contains a warning. Component 5 has the highest median stability but, at a threshold of 0.90, only five stable members out of 361. Its distribution is extremely narrow: 90.6% of its funds lie between 0.85 and 0.90. These are cap-weighted trackers whose weights drift with prices, and the drift ceiling derived above places almost all of them just below 0.90. Figure 2 shows the distribution of ω : it rises to a mode just below 0.90 and falls away sharply above it. A threshold at 0.90 does not select the most passive investors; it selects the investors that rebalance to fixed targets, and these are concentrated in other components.

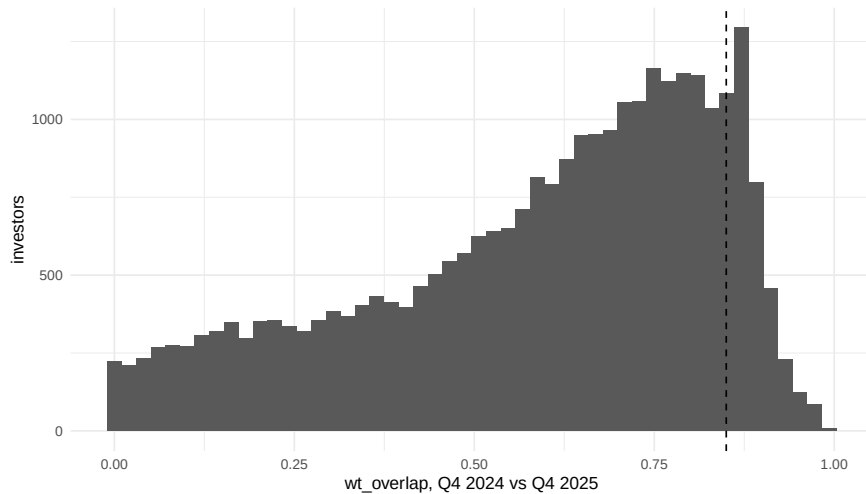


Figure 2: Holdings stability ω between 2024-Q4 and 2025-Q4, 28,709 investors. The dashed line marks the threshold used for `stable_thr`. The mode just below 0.90 is the ceiling that price drift imposes on portfolios that are not traded.

6.2.3 Two anchor rules

Anchors are selected on holdings alone, so that the selection is independent of what the rotation is later used to measure. Two rules are compared:

`stable_thr` investors with $\omega_i(t) \geq \bar{\omega}$. This is the main specification.

`all` every investor present in both years, which is the design of Hamilton et al. (2016) and serves as the benchmark.

Should fewer than 200 investors meet the threshold in a year pair, `stable_thr` falls back to

the 200 most stable investors. Section 6.3.2 explains why the threshold is set to $\bar{\omega} = 0.85$.

6.3 Evaluating the rotations

6.3.1 Design and metrics

No true rotation exists against which an estimate could be scored, so the rules are compared by how the rotations they produce behave on investors they were not fitted to. In every year pair, a random 20% of the common investors (the set E) is removed before either rule selects anchors; both rules are fitted on the remaining 80% and scored on the same E . Five quantities are reported:

- *stable held-out cosine*: the mean cosine between $x_i(t)R$ and $x_i(t+1)$ over members of E with $\omega_i \geq \bar{\omega}$. Investors whose books did not change should land where they were, so higher is better. Because the same threshold defines `stable_thr`, this quantity measures that rule's fit to its own kind of investor and is not an independent test for it;
- *held-out cosine*: the same over all of E ;
- *null*: the held-out cosine obtained when the anchor correspondence is shuffled before fitting, averaged over 20 permutations, which gives the level attainable by chance;
- *effective rank*: the participation ratio $(\sum_k \sigma_k^2)^2 / \sum_k \sigma_k^4$ of the singular values of the anchor matrix, i.e. the number of directions the anchors actually span. A rotation is pinned down only in those directions;
- *median displacement*: the median of $1 - \cos$ over all common investors under the final rotation, which is fitted on the rule's anchors among all common investors.

6.3.2 The threshold

The threshold was initially set at 0.90. At that level only 0.2–5.3% of investors qualify in a given year pair, and in 2006–2007, 2007–2008 and 2008–2009 fewer than 200 do, so `stable_thr` falls back to the 200 most stable investors in exactly the years in which the timing of movement is of most interest. The component-level check in Table 13 exposes the more serious problem. Under the 0.90 rule, the US index trackers, whose books changed less than anyone's, show a median displacement of 0.478, the largest of any component, against 0.186 under `all`. Displacement of that size for near-unchanged portfolios cannot reflect their holdings; it reflects a rotation fitted on anchors that are concentrated elsewhere and extrapolated into a region that holds five of them. The excess displacement under the 0.90 rule follows the number of anchors in a component rather than its turnover, which is what extrapolation error looks like. Across the eight components, the Spearman correlation between median stability and median “stillness” (negative displacement) is 0.74 under `all`, but only 0.10 under the 0.90 rule.

Table 13: Median displacement by Chapter 5 component, 2024–2025, under `stable_thr` at the two thresholds and under the all-investor rule. Anchors are the component’s members meeting the threshold.

Component	n	Anchors		Median displacement		
		≥ 0.90	≥ 0.85	thr 0.90	thr 0.85	all
1	3,974	10	132	0.451	0.369	0.261
2	2,628	12	264	0.316	0.219	0.172
3	4,487	302	785	0.288	0.238	0.216
4	4,842	257	596	0.331	0.357	0.247
5	361	5	332	0.478	0.031	0.186
6	4,003	166	765	0.284	0.221	0.195
7	4,635	178	505	0.345	0.321	0.312
8	3,779	42	199	0.277	0.263	0.215
Spearman(stability, stillness) across components				0.10	0.86	0.74

Lowering the threshold to 0.85 places it below the drift ceiling of buy-and-hold funds. In 2024–2025 the anchor set grows from 972 to 3,578 investors (12.5%), every component contributes at least 132 anchors, and component 5 contributes 332. Its median displacement falls to 0.031 and the component-level correlation rises to 0.86, the highest in the table. In the full panel, the smallest anchor set at 0.85 has 237 investors (2007–2008), so the fallback is never used. The low displacement of component 5 at 0.85 is, however, largely in-sample: 92% of its members are now anchors, and the rotation was fitted to them. The next subsection removes that advantage.

Table 14 reports the five metrics for the 2024–2025 pair at the final threshold. Both rules align far above chance, with held-out cosines of 0.67 and 0.72 against nulls of 0.05 and 0.10. `stable_thr` fits the stable held-out investors better and all held-out investors worse, spans fewer effective dimensions (9.9 against 12.7 for all investors), and implies the higher median displacement. The last point is partly mechanical: among orthogonal maps, the one fitted to all investors is by construction the one that minimises their average displacement, so any other rule must produce more. Which rotation is *correct* cannot be read off this table.

Table 14: Anchor rules for 2024–2025, $\bar{\omega} = 0.85$. Cosines are means over the held-out set; the null shuffles the anchor correspondence. Anchors and median δ refer to the final rotation, which is fitted on the rule’s anchors among all common investors; the three cosine columns come from the held-out design of Section 6.3.

Rule	Anchors	Eff. rank	Stable held-out	Held-out	Null	Median δ
<code>stable_thr</code>	3,578	9.9	0.791	0.673	0.048	0.282
<code>all</code>	28,709	12.7	0.741	0.722	0.104	0.231

Table 15 extends the comparison to all 20 year pairs. The pattern of Table 14 is not particular to the last pair. `stable_thr` attains the higher stable held-out cosine in every one of the 20 pairs, and `all` the higher overall held-out cosine in every one of them. The advantage of `stable_thr` on stable investors grows over the panel, from 0.019–0.034 in the steps up

to 2014 to 0.045–0.055 in 2017, 2020–2023 and 2025, which is what a growing common component in the movement of the investors that trade would produce. The anchor sets are far from uniform in size: `stable_thr` uses between 237 anchors (2007–2008) and 3,578 (2024–2025), and the smallest sets, which occur in the crisis years in which return dispersion pushes even untraded portfolios below the threshold, are also those spanning the fewest effective dimensions. The fallback to the 200 most stable investors is never triggered at $\bar{\omega} = 0.85$. Alignment is far above chance in every pair and under both rules.

Table 15: Anchor rules across the 20 year pairs, $\bar{\omega} = 0.85$. Cosines, effective rank and displacement are medians over pairs.

Rule	Anchors	Eff. rank	Stable h.-o.	Held-out	Null	Median δ
<code>stable_thr</code>	237–3,578	8.6	0.858	0.748	0.087	0.207
<code>all</code>	11,688–28,709	10.1	0.825	0.787	0.152	0.175

Ranges over the 20 pairs: effective rank 4.8–11.5 and 7.1–12.7; stable held-out cosine 0.782–0.930 and 0.739–0.907; held-out cosine 0.669–0.826 and 0.720–0.850.

The effective rank is low for both rules. This is a property of the embeddings: the investor vectors share a strong common direction, which concentrates the singular values. The relevant comparison is therefore between rules, not against the nominal 64 dimensions.

6.3.3 Leave one component out

To compare the rules on a footing where neither has an in-sample advantage, each Chapter 5 component c is removed in turn from the 2024–2025 pair, the two rotations are fitted on the investors of the remaining seven components, and the displacement of c 's members is measured. The comparison is restricted to c 's stable members ($\omega \geq 0.85$), for whom the interpretation is unambiguous: their holdings did not change, so a lower displacement indicates a better rotation. For investors who did trade, a lower displacement is not evidence of a better rotation, because a rotation fitted on everyone partly absorbs movement that is common to many investors.

Table 16: Leave-one-component-out check, 2024–2025. Median displacement of each component's stable members under rotations fitted without that component. The last row weights components by their number of stable members.

Component	Stable members	<code>stable_thr</code>	<code>all</code>
1	132	0.364	0.341
2	264	0.086	0.251
3	785	0.293	0.283
4	596	0.342	0.336
5	332	0.141	0.263
6	765	0.234	0.222
7	505	0.304	0.323
8	199	0.252	0.238
Weighted	3,578	0.261	0.280

Table 16 gives the result. Weighted over all 3,578 stable investors, the stable-anchor rotation places unchanged portfolios closer than the all-investor rotation (0.261 against 0.280), out of sample. The advantage is concentrated rather than uniform. `stable_thr` wins in three components, by a wide margin in component 2 and in component 5, where it places the index trackers at 0.141 against 0.263 under `all` even though none of them is an anchor; it loses by small margins elsewhere. The check supports the departure from the design of Hamilton et al. (2016): anchoring on investors whose holdings did not change yields rotations that place unchanged portfolios more accurately. It is available for one year pair only, because the Chapter 5 labels exist only for 2025.

6.4 Displacement

6.4.1 Definition

Following Hamilton et al. (2016), the displacement of investor i between consecutive years is

$$\delta_i(t) = 1 - \cos(x_i(t) R_{t \rightarrow t+1}, x_i(t+1)), \quad (10)$$

computed under the rotation of each rule. It is zero for an investor that occupies the same position in both years and one for an investor whose two positions are orthogonal. The panel contains 363,459 investor-year pairs, from 11,688 (2008–2009) to 28,709 (2024–2025) per step.

Hamilton et al. (2016) relate displacement to word frequency and polysemy. In the investor setting a more direct validation is available, because the object that drives an investor's position, its portfolio, is observed in both years. If aligned displacement measures something economic, investors that traded more should have moved more.

6.4.2 Displacement tracks turnover

Table 17 reports, for every step, the median turnover and displacement and the Spearman correlation between the two under each rule.

The correlation is positive in all 20 year pairs under both rules, and the 95% interval of every yearly estimate, computed on the Fisher scale with the variance $1.06/(n-3)$ of Fieller et al. (1957), lies above zero. The average of the yearly correlations is 0.49 under `stable_thr` and 0.29 under `all`. Pooling all 363,459 observations gives 0.451 and 0.262 respectively. The pooled figure compares investors across years whose displacement levels differ, and is therefore slightly below the typical yearly value; its nominal interval is too narrow, because the same investors appear in up to 20 steps, and the yearly intervals are the relevant evidence.

Two checks guard against the obvious objections. First, the stayers are the anchors of `stable_thr` and sit close to its rotation by construction, which could produce part of the

Table 17: Displacement and turnover by year pair. Stayers are investors with $\tau \leq 0.15$ (the `stable_thr` anchors). ρ is the Spearman correlation between turnover and displacement.

$t + 1$	n	Median τ	Median δ (<code>stable_thr</code>)			$\rho(\tau, \delta)$	
			All	Stayers	Movers	<code>thr</code>	<code>all</code>
2006	12,237	0.423	0.142	0.046	0.150	0.473	0.347
2007	12,637	0.440	0.157	0.067	0.161	0.482	0.343
2008	11,927	0.441	0.207	0.097	0.209	0.414	0.285
2009	11,688	0.429	0.181	0.102	0.184	0.379	0.235
2010	12,769	0.407	0.175	0.117	0.181	0.388	0.152
2011	13,902	0.386	0.174	0.098	0.182	0.455	0.246
2012	14,506	0.375	0.148	0.085	0.157	0.512	0.347
2013	15,021	0.395	0.178	0.105	0.189	0.472	0.289
2014	16,402	0.385	0.207	0.123	0.218	0.477	0.260
2015	18,391	0.382	0.224	0.119	0.237	0.471	0.330
2016	19,726	0.383	0.243	0.119	0.259	0.527	0.337
2017	18,362	0.384	0.264	0.122	0.284	0.570	0.341
2018	17,238	0.386	0.202	0.114	0.214	0.498	0.371
2019	19,127	0.366	0.204	0.124	0.217	0.504	0.320
2020	21,214	0.413	0.271	0.155	0.281	0.490	0.267
2021	22,755	0.368	0.276	0.165	0.292	0.533	0.281
2022	24,023	0.364	0.253	0.180	0.263	0.513	0.305
2023	25,127	0.347	0.237	0.161	0.251	0.513	0.268
2024	27,698	0.350	0.273	0.197	0.285	0.504	0.275
2025	28,709	0.355	0.282	0.183	0.296	0.528	0.283

correlation mechanically. Excluding them lowers the pooled correlation from 0.451 to 0.408 under `stable_thr`, against 0.289 under `all`. Among investors that traded, the `stable_anchor` rotation therefore links displacement to trading more closely than the `all-investor` rotation. Second, the relationship is monotone rather than driven by a few extreme cases (Table 18). Under `stable_thr`, median displacement rises across every one of the ten turnover deciles, most steeply in the top two, where investors replaced more than about 60% of their top-62 slice. Under `all` it is flat across the lower six deciles and rises only thereafter, which is consistent with a rotation that absorbs the common part of moderate portfolio change.

Table 18: Median displacement by pooled turnover decile (36,346 observations per decile).

Decile	Turnover range	<code>stable_thr</code>	<code>all</code>
1	0.000–0.149	0.136	0.168
2	0.149–0.210	0.178	0.167
3	0.210–0.267	0.192	0.165
4	0.267–0.324	0.199	0.165
5	0.324–0.384	0.209	0.168
6	0.384–0.450	0.218	0.170
7	0.450–0.525	0.231	0.179
8	0.525–0.624	0.257	0.196
9	0.624–0.765	0.312	0.235
10	0.765–1.000	0.449	0.326
Increasing steps		9 of 9	7 of 9

The link is weakest in the steps into 2009 and 2010 under both rules, with the minimum in 2010 (0.388 and 0.152). In the aftermath of the financial crisis, an investor's position in the space appears to have depended more on the reorganisation of holdings around it than on its own trading. Under `stable_thr` the correlation also strengthens over the panel, from about 0.47 in the first steps to about 0.53 in the last.

6.4.3 The level of displacement

Median displacement roughly doubles over the panel, from 0.142 to 0.282, with local peaks in the steps into 2008, 2016–2017 and 2020–2021. Median turnover falls slightly over the same period, from 0.42–0.44 to about 0.35, consistent with the growth of passive vehicles documented in Section 3.5. The displacement of the stayers, investors that traded away at most 15% of their slice, quadruples, from 0.046 to 0.183. Because the stayers are the anchors, their displacement is the in-sample residual of the rotation, a lower bound on alignment error rather than a noise floor, and its growth indicates that consecutive models in the later part of the panel are less well related by a rotation even for unchanged portfolios. The universe more than doubles over the panel, the issuer vocabulary grows with it, and the global mix widens; each of these plausibly makes two independently trained models differ by more than an orthogonal map.

The gap between the median displacement of all investors and that of the stayers is a rough measure of movement beyond this floor. It shows no trend and is largest in the steps into 2008 (0.110), 2015–2017 (0.105–0.142) and 2020–2021 (0.111–0.116). The timing of these peaks is the same under both rules; their size is not. The difference between `stable_thr` and `all` in median displacement is 0.02–0.04 in ordinary years but 0.05–0.06 in 2008, 2017, 2020, 2021 and 2025, so the magnitude of the episodes depends on the anchor choice while their dating does not. The dip in the number of investors between 2016 and 2018 (19,726, 18,362, 17,238), against an otherwise growing panel, suggests a change in source coverage and calls for caution in reading the 2017 step economically.

Figure 3 shows both series. The distance between them, rather than the level of either, carries the timing evidence.

6.5 Do the components persist?

Displacement is continuous and says nothing about the components of Section 5.5. Three complementary procedures address them: a fixed classifier, free per-year refits, and a direct measure of how long investors keep their label. All three start from the Chapter 5 mixture itself, loaded from the stored fit and verified to reproduce the Chapter 5 posterior probabilities to within 3×10^{-4} .

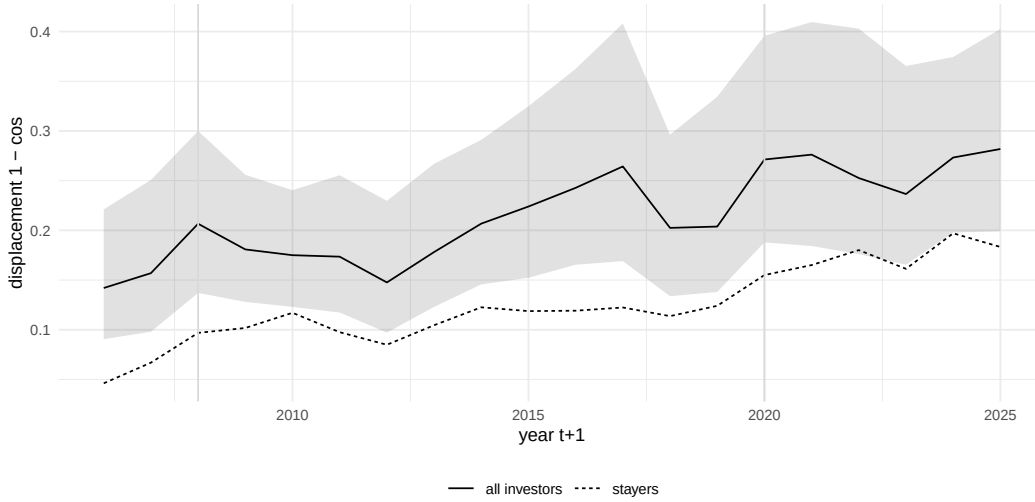


Figure 3: Median displacement by year pair, all investors and stayers, with the interquartile range of all investors shaded. Vertical lines mark the steps into 2008 and 2020.

6.5.1 A fixed classifier

Because the chained rotations of equation (8) map every year into the 2025 frame, the 2025 spherical Cauchy mixture can be applied to each year without re-estimation. Investor i in year t is assigned to the component with the highest posterior probability given $x_i(t)R_{t \rightarrow 2025}$ and the Chapter 5 parameters (π_j, μ_j, ρ_j) . No label matching is required, so transitions between consecutive years are directly comparable.

The assignment depends on the rotation. The labels obtained under `stable_thr` agree with those under `all` for 58.8–88.5% of investors (70.4% on average), with the lowest agreement in 2005–2007. In the early years roughly 40% of investors therefore change component depending only on the anchor rule, which reflects the accumulation of rotation error along the chain. The mean log-likelihood of the aligned years under the 2025 mixture is 12.60 in 2025 itself and between 3.3 and 6.7 in every other year. Part of this gap is the in-sample advantage of the reference year; the rest indicates that aligned earlier years sit more diffusely around the 2025 components than the reference quarter does.

6.5.2 Free refits

In the second procedure, the mixture is re-estimated in every year on the aligned embeddings, with $k = 8$, two restarts and `minprior` set to zero so that all eight components are retained in every year. If the Chapter 5 components are durable populations, each refit should recover them. Refit components are matched to the 2025 components by the Hungarian method (Kuhn, 1955) on the cosine between their location vectors, and the refit partition is compared with the fixed classifier by the adjusted Rand index (ARI) of Hubert and Arabie (1985).

Recovery is weak. The ARI between refit and fixed classifier lies between 0.15 and 0.37 (mean 0.22), and the mean cosine between matched component locations between 0.40 and 0.67 (mean 0.49), with individual components matched at cosines as low as 0.06. The rotation is also a weak guide to which refit component corresponds to which reference component. Matching refit components to the fixed classifier through shared members instead of through their locations yields the same assignment for 12–100% of components, depending on the year. A matching of consecutive refits through the investors they share, which uses no rotation at all, agrees with the location-based matching for 25–100% of components, and consecutive refits agree with each other at ARI 0.15–0.41, declining over the panel.

These values should be read against the reproducibility of the mixture itself. A refit of the 2025 cross-section with the same number of components and restarts but different random initialisations agrees with the Chapter 5 partition at an ARI of only 0.51. The $k = 8$ solution is sensitive to its starting point, a symptom of the flat likelihood surface already visible in the information criteria of Section 5.3. Against that benchmark, the yearly refits recover a moderate part of the Chapter 5 structure, not most of it. Identifying *the same* component across years by re-estimation is therefore unreliable at this resolution, which is why the fixed classifier, which never re-estimates anything, is the primary specification.

6.5.3 Persistence at the level of the investor

The two procedures above ask whether the *partition* is recovered. A weaker and more useful question is whether individual investors stay where they were. Under the fixed classifier, 64.0% of investors present in two consecutive years carry the same label, against 17.8% under independent reassignment with the observed component shares. The share declines slowly with the horizon: averaged over the available starting years it is 61.0% after two years, 53.3% after five and 46.0% after ten, while the chance level is flat at 17.1–17.8% (Table 19). Half of the investors observed a decade apart are still in the same component, roughly three times what chance would give. Membership is therefore persistent even though the components themselves are hard to re-estimate.

Table 19: Share of investors carrying the same component label h years apart, averaged over starting years, against the share expected under independent reassignment.

Horizon h	Starting years	Same component	Chance
1	20	0.640	0.178
2	19	0.610	0.178
3	18	0.576	0.175
5	16	0.533	0.172
7	14	0.503	0.173
10	11	0.460	0.171

Retention differs across components in a way that matches their holdings (Table 20). The US income and real-assets component retains 71.6% of its members from one year to the next

on average, the US mega-cap growth component 69.1% and the concentrated global core 67.3%, while the European quality growth component retains 47.0% and the China mid-cap growth component only 36.3%. Across the seven components the Spearman correlation between median holdings stability (Table 12) and mean retention is 0.61: the components whose members trade least are also those whose membership turns over least. The China component, which Chapter 5 describes as the highest-turnover group and which has the lowest holdings stability in the panel, is also the least persistent, and it is empty or nearly empty in the first two steps.

Table 20: Annual retention by component under the fixed classifier: the share of a component's members in year t still in the same component in year $t + 1$. Component 5 is omitted; see the text.

Component	Mean	Minimum	Maximum	2024–2025
3 US income and real assets	0.716	0.611	0.814	0.616
2 US mega-cap growth	0.691	0.486	0.836	0.685
7 Concentrated global core	0.673	0.504	0.848	0.660
6 Asian technology	0.656	0.508	0.783	0.606
4 Asia–Europe value	0.641	0.455	0.801	0.676
8 European quality growth	0.470	0.163	0.714	0.614
1 China mid-cap growth	0.363	0.000	0.763	0.597

One component is missing from that table, and its absence is a result in itself. Component 5, the US mega-cap index trackers, has no members at all in any aligned year other than the reference quarter: the fixed classifier never assigns an investor to it outside 2025. The cause is its concentration. With $\rho = 0.970$ the spherical Cauchy density falls away from its location vector extremely fast: measured against the uniform density, its value at a cosine of 0.9 is $[(1 - \rho^2)/(1 + \rho^2 - 2\rho \cdot 0.9)]^{63} \approx e^{-75}$, against $\approx e^{264}$ at the location itself. An alignment error far too small to matter for any other component is therefore enough to push a tracker out of the only region in which component 5 is the most likely component, and it is absorbed by a diffuse neighbour. The evidence of Section 6.3.3 shows that these investors are in fact placed accurately by the rotation; what fails is not the alignment but the combination of a very concentrated component with a posterior-based assignment rule. Tracking that population across years would require an assignment rule that is insensitive to small displacements, such as nearest location vector by cosine, or a per-year re-estimation of ρ .

6.6 When do investors move?

Under the fixed classifier, an investor switches in step $t \rightarrow t + 1$ if its component differs between the two years. Table 21 reports the switching rate for all investors, for investors assigned with posterior probability of at least 0.7 in both years, and for the stayers. The chance rate is the switching rate expected if investors were reassigned independently with the observed component shares.

Between 30% and 41% of investors change component from one year to the next. This

Table 21: Component switching under the fixed classifier (`stable_thr`). Excess is the switching rate of all investors minus that of the stayers.

$t + 1$	n	All	Confident	Stayers	Excess	Chance
2006	12,237	0.314	0.263	0.201	0.113	0.785
2007	12,637	0.333	0.282	0.207	0.126	0.777
2008	11,927	0.375	0.334	0.262	0.113	0.789
2009	11,688	0.341	0.296	0.218	0.123	0.802
2010	12,769	0.328	0.278	0.213	0.115	0.817
2011	13,902	0.325	0.265	0.240	0.085	0.822
2012	14,506	0.303	0.246	0.173	0.130	0.817
2013	15,021	0.340	0.287	0.177	0.163	0.818
2014	16,402	0.356	0.304	0.237	0.119	0.823
2015	18,391	0.382	0.334	0.203	0.179	0.833
2016	19,726	0.393	0.339	0.266	0.127	0.839
2017	18,362	0.372	0.315	0.188	0.184	0.828
2018	17,238	0.362	0.306	0.182	0.180	0.813
2019	19,127	0.335	0.277	0.164	0.171	0.815
2020	21,214	0.395	0.341	0.236	0.159	0.828
2021	22,755	0.407	0.361	0.191	0.216	0.837
2022	24,023	0.382	0.331	0.237	0.145	0.833
2023	25,127	0.407	0.368	0.243	0.164	0.853
2024	27,698	0.383	0.340	0.294	0.089	0.857
2025	28,709	0.367	0.328	0.296	0.071	0.856

is far below the 78–86% expected under random reassignment, so the components are not arbitrary labels, but it is far above what durable populations would imply. Restricting to confident assignments lowers the rates only to 25–37%.

The stayers show why raw switching overstates reallocation. Investors that traded away at most 15% of their slice switch component at 16–30%, on average 61% of the rate of all investors. Most switching therefore occurs without trading. With seven of the eight components diffuse (ρ between 0.24 and 0.41 in Section 5.5), many investors lie near a boundary, and the small displacements induced by retraining and alignment are enough to move them across it.

The switching rate varies across steps, though with close to 400,000 transitions and alignment quality that itself varies by step, a formal test of equal step effects would establish variation without identifying its source. The raw series has local peaks in the steps into 2008 (0.375), 2016 (0.393) and 2021 and 2023 (0.407), in line with the dating of the displacement episodes. The excess over the stayers, which removes the part of switching that occurs without trading, does not single out 2008: it lies between 0.09 and 0.13 in 2006–2012, is higher on average in 2013–2023, peaks in the step into 2021 (0.216), and falls to 0.07–0.09 in the last two steps. Only the 2020–2021 episode is visible in both the displacement and the switching evidence.

Switching among the stayers is itself higher at the end of the panel than at the beginning, averaging 0.22 in the steps to 2012 against 0.29–0.30 in the last two, in step with the fourfold

rise in their displacement (Section 6.4).

6.7 Summary and limitations

The chapter transfers the diachronic-embedding design of Hamilton et al. (2016) to investor embeddings and replaces its alignment on the full vocabulary with alignment on investors whose holdings did not change. Five results stand.

First, the choice of anchors matters, and stable anchors are better when chosen with care. A threshold on holdings stability must lie below the ceiling that price drift imposes on buy-and-hold funds; at 0.90 it excludes almost all index trackers and misplaces the most passive component of the market, while at 0.85 it anchors every region. Out of sample, rotations anchored on stable investors place unchanged portfolios more accurately than the all-investor rotation.

Second, aligned displacement tracks portfolio change. Displacement rises with turnover in every year pair, under both anchor rules, without the anchor investors, and monotonically across turnover deciles under `stable_thr`. This is the central validation of the chapter: movement in the embedding space is predominantly movement in the portfolio.

Third, the dating of movement is robust while its level is not. Episodes of elevated displacement in the steps into 2008, 2015–2017 and 2020–2021 appear under both rules; their size depends on the rule, and the secular rise in displacement coincides with a fourfold rise in the displacement of unchanged portfolios.

Fourth, the components are durable as populations of investors but fragile as objects to re-estimate. Under the Chapter 5 mixture applied to the aligned years, 64% of investors keep their component from one year to the next and 46% still share it a decade later, against 17% under chance, and annual retention ranks across components in the same order as their holdings stability ($\rho = 0.61$). A free refit of the same data, by contrast, recovers the reference partition only moderately (ARI 0.15–0.37), against a benchmark of 0.51 for two fits of the same cross-section. The most concentrated component, the index trackers, is not detected at all outside the reference quarter, not because its members are misplaced by the rotation but because a component with $\rho = 0.97$ cannot absorb even a small alignment error under a posterior-based assignment rule.

Fifth, a substantial part of measured movement is not reallocation. Between 30% and 41% of investors change component each year, far below the 78–86% expected under chance, but investors that traded away at most 15% of their slice switch at 16–30%, so most switching occurs without trading. Switching among those same investors drifts upward over the panel in step with the displacement of unchanged portfolios. Read against that floor, the episodes in which investors genuinely change position are the steps into 2008, 2015–2017 and, most clearly, 2020–2021.

The out-of-sample comparison of anchor rules is available only for the year pair ending in

the reference quarter. The stability measure is defined on the 62-position slice the encoder reads, so trading outside that window is invisible to it by design. And the posterior-based assignment rule is not suited to very concentrated components: component 5 is untrackable under it, so nothing in this chapter speaks to the persistence of the passive population, which the displacement evidence suggests is the most stable of all.

Taken together, these results answer the second half of the research question of Section 1.3 with a qualification. The components recovered in one cross-section are not an artefact of that cross-section: investors keep their membership far longer than chance, and the components that hold the most stable portfolios are also the ones whose membership turns over least. Investors do move between them at identifiable moments, and the dating of those moments agrees across the displacement and switching evidence for 2020–2021 and, more weakly, for 2008 and 2015–2017. What the chapter cannot support is the stronger reading that each component is a fixed object whose boundaries can be re-estimated year by year. At $k = 8$ the partition is too dependent on initialisation for that, and the classification is best used as Chapter 5 uses it: as a description of one cross-section, whose populations, this chapter now shows, persist.

7 Discussion and Conclusion

This thesis asked whether investor embeddings learned from portfolio holdings alone recover groups of investors that are economically distinct, and whether those groups are stable over time. They do, but what makes them distinct is mostly the investable universe rather than investment orientation; and they are durable as populations of investors while being fragile as objects to re-estimate.

7.1 What replicates, and what the extensions add

The investor-side pipeline of Gabaix et al. (2025) reproduces on a different sample without difficulty: the filters, sequence construction, encoder and two-stage training are taken verbatim, and the models behave as described. The resulting embeddings carry information never given to the encoder—an investor’s ten nearest neighbours share its declared style 2.7 times more often than chance, its domicile 4.7 times more often—and they isolate the US mega-cap index trackers as a single component at $\rho = 0.970$ under every specification tried, with distinctive holdings held by 85–98% of members. When a group genuinely holds near-identical portfolios, the method finds it.

Four of these extensions carry the argument beyond that. First, aggregation is not innocuous. The two arms differ only in the read-out, and weighting a position by the capital committed to it reorders the association ranking: breadth falls from 0.613 to 0.521 and capitalisation rises from 0.276 to 0.386, overtaking geography (Section 5.4). Second, replacing neighbourhood inspection with a mixture makes the classification executable and the scoring procedure of Section 5.5.6 makes it falsifiable, at the cost of exposing how little the likelihood constrains the number of components (Section 5.3). Third, the breadth of the universe decides what the geometry encodes: globally the components are organised by portfolio width, capitalisation and geography, with style fifth, while restricting to investors at least 90% invested in US issuers lifts the strategy family from 0.201 to 0.364 and capitalisation to 0.743 (Section 5.7). Investment orientation is a within-segment phenomenon, which reconciles these results with the strategy-like groupings the source paper reports on a US-only universe. Fourth, the classification survives outside its own cross-section: across twenty aligned annual steps, 64% of investors keep their component from one year to the next and 46% still share it a decade later, against 17% under chance, and retention ranks across components in the same order as the stability of their holdings (Section 6.5).

7.2 Implications

Section 2.2 identified a common feature of the herding, crowding and style literatures: each needs a prior judgement about which investors are alike, and each takes it from an institutional type, a declared style or a factor exposure, all exogenous to the holdings they interpret. Constructing that judgement from the holdings themselves yields a caution. On

a global sample, learned similarity groups investors by what they are able to hold, so two funds judged alike may share a mandate region and a capitalisation band without sharing a bet. A crowding measure built on such a grouping would flag investors subject to the same eligibility constraints rather than those holding the same position. The remedy is visible in Section 5.7: holdings-based similarity is informative within a segment rather than across segments.

Declared labels come out neither vindicated nor dismissed. Style scores 0.201 against component membership, well behind breadth, manager and geography, yet it agrees with the independently measured factor profile wherever the two can be compared: the income component carries a Yield label and the highest dividend yield in the panel, the growth component a Growth label and the highest beta and price-to-book. Labels describe something real.

For the temporal gap, the contribution is a method and a calibration. Investor populations recovered from holdings persist over a decade far beyond chance. But a large share of measured movement between components occurs among investors that did not trade, so any dynamic use of learned representations needs a floor of that kind before its episodes can be believed.

7.3 Limitations and further work

Three limitations bear on how the results should be read. Truncation is confounded with the leading result: the encoder reads at most 62 positions, 42.6% of investor-quarters exceed that window, and portfolio breadth is both the strongest attribute in the ranking and the one most entangled with the input construction. The US-only comparison is post hoc, so geography is removed from the clustering but not from the representation. The declared attributes are current snapshots rather than histories (Section 3.4), which is why the dynamic analysis relies on holdings-derived quantities.

Each points to an extension. Reading the whole portfolio—by aggregating across chunks or enlarging the context window—would let breadth enter the representation as a property of the investor rather than of the input. A magnitude-aware objective, in which dollar holdings enter what the model predicts rather than only the read-out, completes the weighted-pooling extension and is the refinement Gabaix et al. (2025) name. Two applications on their list also remain: performance evaluation, which the persistence results now make a coherent question, and crowding detection, which the findings here say must be conducted within an investable universe. Finally, a component as concentrated as the index trackers cannot absorb even a small alignment error under a posterior-based assignment rule; assigning by nearest location vector would recover the one population whose persistence the present design cannot measure.

7.4 Conclusion

Portfolio holdings, read as sequences and compressed by a transformer encoder, support a classification of institutional investors built without any prior vocabulary and given economic content afterwards. On a global sample that classification is organised principally by the investable universe—what an investor is able to hold—with investment orientation a within-segment phenomenon that becomes legible once the universe is held fixed. The groups are not artefacts of the quarter in which they are estimated: investors keep their membership far longer than chance over a twenty-year panel, and the groups whose members trade least are those whose membership turns over least. What the method does not support is treating those groups as fixed objects with re-estimable boundaries. Within those limits, the investor side of the asset-embeddings framework sustains the application for which its authors set it aside.

A Implementation Notes

A.1 Software and reproducibility

Data preparation and all statistical analysis are implemented in R 4.5.1 on aarch64-apple-darwin20; embedding models are trained in Python (PyTorch) on a single NVIDIA A30 GPU on the WU research cluster, one model per quarter. Except for the BIC scan (which uses seed 4525), every stage uses the fixed random seed 42: the per-quarter encoder training, the train/validation split, the EM initialisation of the mixture, and the subsampling used for the internal validity indices. Package versions are listed in Table 22. Cluster-level analysis is consolidated in a single script that reads cached inputs and selects the specification through one configuration switch, so that the replicated and weighted arms are produced by identical code.

The full pipeline—WRDS extraction, filtering, tokenisation, per-quarter training, clustering, characterisation and the dynamics analysis—is available at

<https://github.com/Jack00027/Master-s-Thesis>,

together with the scripts that produce every table and figure reported here. The holdings data are licensed from FactSet through WRDS and cannot be redistributed, so the repository contains code and configuration only.

Table 22: R package versions used for the reported results.

Package	Version	Package	Version
arrow	21.0.0.1	circlus	0.0.2
dplyr	1.1.4	skmeans	0.2-18
tidyr	1.3.1	cluster	2.1.8.1
purrr	1.1.0	RPostgres	1.4.8
flexmix	2.3-20	dbplyr	2.5.1

B Supplementary Tables

This appendix reports in full the descriptive tables summarised in Section 5.5. All figures are for the weighted arm on the global universe at 2025-Q4 with $k = 8$; component numbering and labels follow Table 8. Categorical descriptors are reported as *lift*—the share of a category within a component divided by its share in the population—as set out in Section 5.5.6, and only the categories with the highest lift in each component are listed, which is why the tables are not exhaustive partitions of each attribute.

Table 23: Specification of the reported run.

Quarter	2025-Q4 (31 December 2025)
Arm	weighted (position weights = portfolio share)
Universe	global
Components k	8
BIC at the reported fit	−810,980
minprior	0.01 (floor of 324 investors)
Smallest component	410 investors
Investors	32,389
Positions in the top-62 slice	1,623,786
Investors matched in the reference table	99.9 %
Investors with a declared style	99.0 %
Issuers resolved to a name	100.0 %

Notes: The top-62 slice is joined across chunks; the first chunk alone would have been shorter than the slice for 14,021 of 32,389 investors (median 50 positions against 62). Eight issuers carry no name in the security table, the most widely held of them appearing in 203 portfolios (0.6 % of investors), so the holdings tables below are effectively complete.

B.1 Who is in each component

Table 24: Investor type by component: the two most over-represented types in each. OEF open-end fund, ETF exchange-traded fund, CEF closed-end fund, VAR variable annuity fund, HF hedge fund.

Component	Type	<i>n</i>	Share (%)	Lift
1	OEF	3,285	77.1	1.08
1	VAR	128	3.0	0.97
2	VAR	161	5.3	1.72
2	CEF	49	1.6	1.33
3	VAR	240	4.8	1.56
3	ETF	1,512	30.5	1.40
4	CEF	76	1.4	1.15
4	OEF	3,982	73.8	1.03
5	VAR	54	13.2	4.25
5	ETF	177	43.2	1.98
6	ETF	1,587	35.3	1.62
6	VAR	214	4.8	1.53
7	HF	270	4.8	2.07
7	CEF	73	1.3	1.06
8	OEF	3,598	85.1	1.19
8	CEF	48	1.1	0.93

Notes: The fund-type classification (*fund_type_desc*) is a near-relabelling of the investor type for the registered vehicles and reproduces this ordering to within 0.1 in lift; it is not tabulated separately.

Table 25: Declared style by component: distribution within each component (%). Rows sum to 100 over investors carrying a style label (99.0 % of the sample).

Comp.	Agg. Growth	Deep Value	GARP	Growth	Index	Value	Yield
1	40.5	0.8	3.4	22.1	22.5	7.0	3.7
2	15.8	0.1	5.6	55.6	15.7	5.5	1.7
3	10.1	3.2	6.3	18.2	29.1	13.2	19.9
4	9.8	1.8	18.2	23.2	28.6	10.3	8.1
5	0.2	0.0	0.0	25.4	74.4	0.0	0.0
6	29.5	0.4	8.4	20.0	38.2	2.4	1.1
7	31.0	3.1	9.7	21.6	15.8	9.0	9.7
8	28.6	1.6	7.1	38.2	7.9	9.5	7.0

Table 26: Declared style by component: the three most over-represented styles in each, by lift. Distributions are in Table 25.

Component	Style	<i>n</i>	Share (%)	Lift
1	Aggressive Growth	1,693	40.5	1.75
1	Index	941	22.5	0.95
1	Value	291	7.0	0.84
2	Growth	1,661	55.6	2.09
2	Aggressive Growth	473	15.8	0.68
2	Index	469	15.7	0.66
3	Yield	980	19.9	2.55
3	Deep Value	156	3.2	1.84
3	Value	652	13.2	1.59
4	GARP	977	18.2	2.08
4	Value	553	10.3	1.24
4	Index	1,535	28.6	1.21
5	Index	305	74.4	3.14
5	Growth	104	25.4	0.95
6	Index	1,713	38.2	1.61
6	Aggressive Growth	1,321	29.5	1.27
6	GARP	375	8.4	0.96
7	Deep Value	171	3.1	1.81
7	Aggressive Growth	1,701	31.0	1.34
7	Yield	534	9.7	1.25
8	Growth	1,613	38.2	1.44
8	Aggressive Growth	1,207	28.6	1.24
8	Value	401	9.5	1.14

Table 27: Declared turnover class by component: the two most over-represented classes in each.

Component	Turnover class	<i>n</i>	Share (%)	Lift
1	High	935	22.0	2.43
1	Very High	78	1.8	1.74
2	Low	681	22.6	1.30
2	Very Low	1,841	61.0	1.08
3	Very Low	3,426	69.2	1.22
3	Low	764	15.4	0.89
4	Very Low	3,236	60.0	1.06
4	High	485	9.0	0.99
5	Very Low	392	95.6	1.69
6	Very Low	2,971	66.1	1.17
6	Low	721	16.0	0.92
7	Very High	96	1.7	1.62
7	High	750	13.4	1.47
8	Low	938	22.2	1.28
8	Medium	693	16.4	1.03

Table 28: Investor domicile by component: the largest domicile in each. Domicile is where the vehicle is registered, not where it invests; the Luxembourg and Irish shares reflect the UCITS wrapper rather than the location of the manager.

Component	Domicile	Share (%)
1	China	57.4
2	United States	24.5
3	United States	35.2
4	United States	21.9
5	United States	59.1
6	United States	29.6
7	China	19.6
8	Luxembourg	23.2

Table 29: Investor domicile by component: the three most over-represented domiciles in each, by lift.

Component	Domicile	<i>n</i>	Share (%)	Lift
1	Israel	371	10.6	4.36
1	Brazil	158	4.5	3.87
1	China	2,002	57.4	3.01
2	Netherlands	36	1.5	4.10
2	Denmark	48	2.0	3.69
2	Italy	38	1.6	3.40
3	Japan	111	2.7	4.50
3	Norway	48	1.2	3.49
3	Sweden	152	3.7	3.00
4	Thailand	286	6.6	4.44
4	Poland	22	0.5	4.33
4	India	548	12.7	3.71
5	United States	194	59.1	2.43
5	Ireland	32	9.8	1.70
5	Canada	20	6.1	1.15
6	Hong Kong	69	1.9	3.28
6	South Korea	89	2.4	2.55
6	India	223	6.1	1.78
7	Chile	28	0.6	4.79
7	Taiwan	254	5.8	3.80
7	Mexico	62	1.4	3.60
8	France	222	6.3	2.76
8	Spain	214	6.1	2.24
8	Liechtenstein	28	0.8	2.22

Notes: Coverage is 81.2% of investors; shares are computed over investors with a recorded domicile.

B.2 How they invest

Table 30: Portfolio characteristics by component: medians of the investor-level characteristics supplied by FactSet.

Comp.	<i>n</i>	Beta	Price- earnings	Price- book	Dividend yield	Sales growth	Price momentum	Relative strength
1	4,258	0.68	31.9	3.59	0.53	9.63	0.57	0.54
2	3,018	0.92	29.1	6.51	0.61	11.90	0.76	0.68
3	4,956	0.51	25.6	3.63	0.77	7.25	0.59	0.50
4	5,396	0.32	25.3	3.56	0.57	5.28	0.42	0.38
5	410	1.16	31.2	7.22	0.60	15.80	0.93	0.89
6	4,498	0.54	30.5	4.20	0.48	8.14	0.53	0.51
7	5,625	0.59	27.9	3.96	0.50	8.06	0.55	0.48
8	4,228	0.57	26.9	4.60	0.52	7.30	0.53	0.47

Notes: Coverage is 92.5–93.4 % depending on the field. The characteristics are current snapshots, as discussed in Section 3.4.

Table 31: Size and breadth by component. Assets under management are in the units supplied by FactSet (panel median 113,586, maximum 11,600,000) and are reported only for the investors for which the field is populated. Median positions counts the full portfolio, not the top-62 slice; truncated is the share of members whose portfolio exceeds the 62-position window read by the encoder.

Component	Median AUM (raw units)	AUM known (%)	Median positions	Truncated (%)
1	220,419	37.5	87	78.1
2	277,194	44.0	168	94.4
3	162,887	53.5	73	59.9
4	132,572	46.1	63	50.2
5	402,994	60.2	500	100.0
6	177,449	54.5	70	56.2
7	25,248	38.2	29	0.3
8	54,452	44.0	46	3.1

B.3 What they hold

Table 32: Issuer country by component: the largest country of issue in each. Shares are of positions in the top-62 slice, weighted by portfolio share.

Component	Country	Share of positions (%)
1	China	59.0
2	United States	67.1
3	United States	42.7
4	China	21.0
5	United States	99.2
6	United States	31.7
7	United States	32.3
8	United States	31.6

Table 33: Issuer country by component: the three most over-represented countries in each, by lift.

Component	Country	Positions	Share (%)	Lift
1	Israel	23,327	9.6	4.86
1	Brazil	13,497	5.3	3.31
1	China	149,193	59.0	2.67
2	Ireland	5,056	4.8	3.00
2	Luxembourg	589	0.7	2.12
2	United States	116,985	67.1	2.00
3	New Zealand	652	0.3	4.63
3	Norway	2,592	1.1	3.93
3	Australia	8,955	3.5	3.86
4	Malta	59	0.0	5.40
4	Thailand	15,981	6.4	4.23
4	Lithuania	52	0.0	3.76
5	United States	24,813	99.2	2.96
5	United Kingdom	389	0.5	0.13
5	Ireland	156	0.2	0.12
6	Saudi Arabia	1,384	0.4	4.06
6	Hungary	391	0.1	2.97
6	South Korea	11,120	4.4	2.93
7	Chile	646	0.4	4.09
7	Turkey	262	0.2	3.25
7	Mexico	1,504	1.0	2.93
8	Argentina	87	0.0	3.15
8	Indonesia	970	0.4	3.08
8	Uruguay	535	0.3	2.68

Notes: For component 5 the two non-US entries are reported because they are the only non-US countries it holds at all; their lifts are far below one and identify the component by what it does *not* hold.

Table 34: Issuer region mix: share of positions by FactSet region code (%).

Component	A	E	F	L	M	N	Y
1	60.6	2.5	0.1	5.4	9.7	21.6	0.1
2	4.4	25.4	0.0	0.1	0.5	69.2	0.2
3	21.3	21.6	0.0	0.2	0.2	53.0	3.8
4	52.0	26.3	2.3	0.2	0.9	17.5	0.8
5	0.0	0.7	0.0	0.0	0.0	99.3	0.0
6	50.0	13.5	0.9	1.4	0.8	33.1	0.3
7	34.7	19.6	1.2	4.1	2.9	36.8	0.6
8	26.3	35.9	0.5	1.4	0.6	34.7	0.6

Notes: Region codes are reported as supplied by the vendor. Reading them against the country composition of each component in Table 33, A corresponds to Asia, E to Europe, F to Africa, L to Latin America, M to the Middle East, N to North America and Y to Australasia; the mapping is inferred rather than documented in the extract.

Table 35: Capitalisation profile by component: share of positions by capitalisation group (%) and the median market capitalisation of the positions held.

Component	Mega	Large	Mid	Small	Micro	Median cap (USD m)	Cap known (%)
1	17.8	34.4	35.4	11.8	0.7	6,298	99.6
2	84.5	11.8	2.9	0.5	0.4	172,623	99.1
3	39.7	34.3	19.5	6.2	0.3	21,104	99.8
4	28.7	24.2	28.4	17.1	1.7	6,566	97.5
5	99.9	0.1	0.0	0.0	0.0	292,563	99.9
6	62.4	29.4	7.1	1.1	0.1	35,498	99.8
7	38.7	29.4	21.8	9.3	0.7	18,565	99.0
8	52.2	28.5	15.1	4.0	0.2	38,231	99.7

Table 36: Industry by component: the three most over-represented FactSet industries in each, by lift on positions.

Component	Industry	Positions	Share (%)	Lift
1	Pharmaceuticals: Generic	363	0.3	3.55
1	Textiles	1,257	0.4	3.02
1	Aluminum	2,245	1.0	2.96
2	Telecommunications Equipment	3,361	5.6	2.55
2	Investment Trusts/Mutual Funds	3,603	6.7	2.36
2	Home Improvement Chains	1,502	0.5	2.16
3	Real Estate Investment Trusts	24,434	10.1	4.75
3	Oil & Gas Pipelines	2,725	1.2	3.16
3	Railroads	1,981	0.8	2.80
4	Publishing: Books/Magazines	597	0.2	3.08
4	Hospital/Nursing Management	2,494	0.8	2.56
4	Office Equipment/Supplies	149	0.0	2.48
5	Telecommunications Equipment	410	10.7	4.92
5	Home Improvement Chains	421	0.9	3.97
5	Property/Casualty Insurance	390	2.1	3.67
6	Internet Retail	4,849	3.7	2.03
6	Oil Refining/Marketing	1,538	1.0	2.02
6	Semiconductors	18,032	15.5	1.91
7	Contract Drilling	310	0.2	1.94
7	Oil & Gas Pipelines	891	0.7	1.85
7	Pulp & Paper	326	0.2	1.68
8	Consumer Sundries	229	0.1	2.19
8	Apparel/Footwear	2,844	1.4	2.01
8	Environmental Services	664	0.4	1.99

Table 37: Distinctive holdings, components 1–4: issuers held by at least 10 % of a component's members, ranked by lift. Share is the fraction of the component's members holding the issuer in their top-62 slice; lift is that share divided by the corresponding share in the population.

Component	Issuer	Share of members	Lift
1	Sinomine Resource Group Co., Ltd.	0.11	5.6
1	Giant Network Group Co. Ltd.	0.10	5.2
1	EVE Energy Co., Ltd.	0.10	4.6
1	Jiangsu Hengli Hydraulic Co., Ltd.	0.10	4.6
1	Zhejiang Huayou Cobalt Co., Ltd.	0.13	4.4
1	Shengyi Technology Co., Ltd.	0.14	4.2
1	XCMG Construction Machinery Co., Ltd.	0.11	4.2
1	Suzhou Dongshan Precision Manufacturing Co., Ltd.	0.11	4.1
2	JPMorgan Chase & Co.	0.68	5.3
2	Bank of America Corp.	0.48	5.1
2	Walmart, Inc.	0.44	5.1
2	Morgan Stanley	0.36	5.1
2	AbbVie, Inc.	0.50	5.0
2	General Electric Co.	0.32	4.9
2	Eli Lilly & Co.	0.69	4.9
2	Tesla, Inc.	0.53	4.8
3	Prologis, Inc.	0.11	3.0
3	Merck & Co., Inc.	0.13	1.3
3	Cisco Systems, Inc.	0.12	1.2
3	Johnson & Johnson	0.13	1.2
3	JPMorgan Chase & Co.	0.10	0.8
3	Alphabet, Inc.	0.16	0.7
3	Microsoft Corp.	0.15	0.6
3	Amazon.com, Inc.	0.11	0.5
4	GSK Plc	0.10	2.2
4	Prosus NV	0.12	2.2
4	Münchener Rückversicherungs-Gesellschaft AG	0.10	2.1
4	Banco Bilbao Vizcaya Argentaria SA	0.11	2.1
4	Intesa Sanpaolo SpA	0.11	2.1
4	UniCredit SpA	0.12	2.1
4	BNP Paribas SA	0.11	2.0
4	ING Groep NV	0.10	1.9

Table 38: Distinctive holdings, components 5–8. Definitions as in Table 37.

Component	Issuer	Share of members	Lift
5	Raytheon Technologies Corp.	0.87	18.5
5	AT&T, Inc.	0.93	18.4
5	The Boeing Co.	0.58	18.3
5	GE Vernova, Inc.	0.90	17.3
5	Verizon Communications, Inc.	0.90	16.9
5	Philip Morris International, Inc.	0.85	16.8
5	Amgen, Inc.	0.93	16.4
5	McDonald's Corp.	0.98	16.4
6	ASE Technology Holding Co., Ltd.	0.10	5.2
6	KB Financial Group, Inc.	0.12	4.9
6	PDD Holdings, Inc.	0.13	4.7
6	Hon Hai Precision Industry Co., Ltd.	0.12	4.6
6	NetEase, Inc.	0.16	4.3
6	Delta Electronics, Inc.	0.14	4.2
6	MediaTek, Inc.	0.15	4.2
6	Xiaomi Corp.	0.14	4.2
7	Amazon.com, Inc.	0.18	0.9
7	Alphabet, Inc.	0.21	0.9
7	Microsoft Corp.	0.21	0.8
7	Taiwan Semiconductor Manufacturing Co., Ltd.	0.13	0.8
7	NVIDIA Corp.	0.16	0.7
7	Meta Platforms, Inc.	0.13	0.7
7	Apple, Inc.	0.12	0.6
7	ASML Holding NV	0.10	0.6
8	Compagnie de Saint-Gobain SA	0.11	2.7
8	Infineon Technologies AG	0.12	2.4
8	Novo Nordisk A/S	0.19	2.4
8	Schneider Electric SE	0.25	2.4
8	AIA Group Ltd.	0.13	2.4
8	L'Oréal SA	0.15	2.3
8	MercadoLibre, Inc.	0.12	2.3
8	RELX Plc	0.10	2.2

Notes: Issuer names are as recorded in the security table, lightly tidied for capitalisation. Components 3 and 7 have no issuer held at a lift materially above one; the entries listed for them are their most widely held names, which they hold less often than the population, and this is the substantive point made in Section 5.5.

B.4 Administrative structure and declared mandate

Table 39: Manager concentration by component. A component dominated by a single asset manager would be a product lineup rather than an investment grouping; none is.

Component	Largest managing institution	Share of component (%)	Distinct managers	Top three managers (%)
1	E Fund Management Co., Ltd.	2.9	576	8.5
2	Fidelity Management & Research Co. LLC	2.2	667	5.6
3	BlackRock Fund Advisors	2.0	1,050	4.9
4	Fidelity Management & Research Co. LLC	3.6	1,003	6.3
5	SSgA Funds Management, Inc.	4.3	156	11.8
6	Amundi Asset Management SA	2.2	752	6.0
7	Amundi Asset Management SA	1.3	1,483	2.8
8	Schroder Investment Management Ltd.	1.0	1,195	2.9

Notes: Amundi is recorded in the entity master as “Amundi Asset Management SA (Investment Management)”.

Table 40: Declared mandate region by component: the two most over-represented regions in each, by FactSet code.

Component	Region code	<i>n</i>	Share (%)	Lift
1	Y	97	2.6	3.14
1	Z	37	1.0	2.11
2	G	2,108	79.3	1.87
2	N	49	1.8	1.54
3	N	75	1.9	1.63
3	X	83	2.2	1.22
4	X	176	4.0	2.28
4	E	392	9.0	1.81
5	N	40	13.8	11.60
5	D	197	67.9	1.58
6	R	593	16.7	4.74
6	A	101	2.8	1.76
7	Z	31	0.7	1.47
7	G	2,091	47.1	1.11
8	A	141	3.9	2.39
8	E	346	9.5	1.92

Notes: Codes are reported as supplied by FactSet; coverage is 81.8% of investors.

Table 41: Declared specialisation by component: the two most over-represented specialisations in each, by FactSet code.

Component	Specialisation code	n	Share (%)	Lift
1	F001	285	7.5	2.77
1	M004	202	5.3	2.65
2	M002	58	2.0	3.81
2	M003	62	2.2	2.39
3	E012	318	7.8	2.80
3	E003	170	4.2	2.31
4	O006	41	0.8	4.65
4	E002	495	10.1	2.53
5	E004	143	35.2	6.06
5	E010	72	17.7	2.48
6	E007	363	9.3	2.04
6	E004	423	10.9	1.87
7	F001	209	4.5	1.66
7	O009	48	1.0	1.51
8	E011	57	1.5	2.67
8	E001	1,296	34.5	1.38

Notes: Codes are reported as supplied by FactSet; coverage is 87.7 % of investors.

B.5 The association ranking in full

Table 42 reports every attribute scored, extending Table 9 with the number of categories entering each contingency table. Cardinality is reported because the bias correction of Bergsma (2013) removes the expected inflation of Cramér's V under independence but does not make high- and low-cardinality attributes fully commensurable, as discussed in Section 5.5.6.

Table 42: Attributes ranked by association with component membership, full listing. Categorical attributes are scored by bias-corrected Cramér's V , continuous ones by rank ϵ^2 . Levels is the number of categories; coverage is the share of investors for which the attribute is observed.

Attribute	Kind	Strength	Levels	Coverage	Family
n_assets_full	continuous	0.521	—	1.000	breadth
mgr_entity_proper_name	categorical	0.386	2,823	0.957	artifact
med_cap	continuous	0.386	—	0.995	size
modal_issuer_country	categorical	0.316	46	1.000	geography
inv_country_desc	categorical	0.293	58	0.812	geography
fs_ultimate_parent_entity	categorical	0.288	4,161	0.813	artifact
inv_obj_region_code	categorical	0.210	13	0.818	geography
style_any	categorical	0.201	7	0.990	strategy
inv_obj_specialization_code	categorical	0.192	51	0.877	sector
pct_us	continuous	0.164	—	1.000	geography
beta	continuous	0.159	—	0.933	strategy
turnover_any	categorical	0.150	5	0.999	strategy
fund_type_desc	categorical	0.140	4	0.977	vehicle
pb_ratio	continuous	0.140	—	0.931	strategy
investor_type	categorical	0.127	5	1.000	vehicle
inv_obj_asset_type_code	categorical	0.127	6	0.876	vehicle
sales_growth	continuous	0.126	—	0.934	strategy
price_momentum	continuous	0.120	—	0.934	strategy
relative_strength	continuous	0.104	—	0.925	strategy
aum_any	continuous	0.092	—	0.456	size
pct_top_sec	continuous	0.064	—	1.000	sector
pe_ratio	continuous	0.036	—	0.932	strategy
dividend_yield	continuous	0.035	—	0.934	strategy

Notes: inv_country_desc is where the vehicle is registered, modal_issuer_country where it invests. Within the strategy family the strongest attribute is the declared style (0.201) and the second is beta (0.159).

Table 43: Sensitivity of the holdings-level scores to the aggregation rule. Attributes measured at the holdings level are aggregated to the investor either by portfolio weight or by position count; the weighted versions are reported throughout the thesis.

Attribute	ϵ^2 , weighted	ϵ^2 , count-based
pct_top_sec	0.064	0.067
med_cap	0.386	0.375

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